

**COMPUTATIONAL METAGENOMICS ANALYSIS OF ONE HEALTH-
ORIENTED ANTIMICROBIAL RESISTANCE DYNAMICS OF
WASTEWATER ACROSS FIVE MAJOR CITIES**

Submitted By

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Under the Supervision of

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BONAFIDE CERTIFICATE

This is to certify that this project entitled "**Computational Metagenomics analysis of One health-oriented Antimicrobial Resistance Dynamics of Wastewater across Five major cities**" is a Bonafide record of work done by **PIYUSH MESHRAM**, Reg. No. **24MSBINPY0036**, pursuing Master of Science in Bioinformatics at Pondicherry University, Pondicherry, and has been carried out under my direct supervision and guidance.

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DECLARATION

I hereby declare that the dissertation entitled " **Computational Metagenomics analysis of One health-oriented Antimicrobial Resistance Dynamics of Wastewater across Five major cities** " is submitted for partial fulfilment of completion of my project for Master of Science in Bioinformatics. This work is entirely original, produced by me at the Department of Bioinformatics, Pondicherry University, Pondicherry. This work has not been submitted elsewhere for any other degree, diploma, associateship or fellowship.

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ABSTRACT

Antimicrobial resistance (AMR) is a critical global health crisis requiring advanced monitoring systems to track its environmental spread. Wastewater-based epidemiology has developed into a strong monitoring solution which provides real-time tracking of population-level resistance through its One Health approach. This thesis investigates the "functional resistance profiles" of urban wastewater across five Indian hubs, comparing Tier 1 megacities (Delhi and Kolkata) with transitioning Tier 2 cities (Jaipur, Vellore, and Cuttack).

The research examined 29 extensive metagenomic datasets that were obtained from the NCBI Sequence Read Archive. The bioinformatic pipeline started with essential quality control procedures before it performed human DNA removal and implemented a "direct mapping" method. The researchers used DIAMOND to find resistance genes which they matched against the Comprehensive Antibiotic Resistance Database (CARD) while Centrifuge delivered taxonomic identification of the related microbial populations.

The study results demonstrate that urbanization intensity determines how functional resistance profile structure develops. The "core" urban resistance profile which shows consistent patterns throughout India maintains its stability yet megacities function as active centres for clinical development. The cities of Delhi and Kolkata functioned as urban breeding grounds for superbugs because they accumulated dangerous quantities of high-risk carbapenemase genes (*bla*NDM, *bla*OXA) and Extended-Spectrum Beta-Lactamases (ESBLs).

The study of the Kathajodi River in Cuttack showed that upstream water maintained its natural state while downstream reservoirs showed extreme urban pollution which resulted in taxonomic identification of resistance spikes to *Klebsiella pneumoniae* and *Acinetobacter baumannii*. The treatment analysis revealed that treated effluent contained substantial "diversity residues," which enabled multiple types of ARGs to flow into the environmental water column.

The research demonstrates how wastewater surveillance can deliver essential genomic data which assists national AMR monitoring programs. The optimized, computationally efficient pipeline developed here offers a scientifically rigorous blueprint for AMR monitoring in resource-limited regions where the public health need is urgent. This study promotes advanced tertiary treatment standards to eliminate genetic diversity from urban functional resistance profiles through its integration of bioinformatics with environmental policy.

CHAPTER 1: INTRODUCTION

1.1 Defining Antimicrobial Resistance: A Global Biological Challenge

Antimicrobial Resistance (AMR) represents an evolutionary process which enables microorganisms to develop resistance against all antimicrobial substances that include antibiotics, antifungals and antivirals. Through genetic mutations bacteria become resistant to drugs because of their natural ability to develop resistance which human activities have accelerated through their frequent and improper use of antimicrobial drugs in all medical fields and veterinary practices and agricultural operations (WHO, 2024).

Bacteria develop resistance through multiple functional pathways which include the following mechanisms (Munita & Arias, 2016):

- 1 **Enzymatic Inactivation** Through their production of enzymes bacteria create drugs which they use to destroy or change existing medications (for example beta-lactamases).
- 2 The process known as **efflux pumps** enables cells to utilize their internal systems to push antimicrobial substances out of their bodies before those substances can affect their targets.
- 3 The process of **target modification** enables organisms to create structural changes in their cell components which drugs use as their target sites.
- 4 The process of **reduced permeability** enables bacteria to block drug entry by creating thicker cell walls which they modify.

The emergence of resistant strains which medical professionals label as superbugs (Cleveland Clinic, 2024; Paul et al., 2025) causes standard medical treatment methods to lose their effectiveness. Patients experience longer sickness durations which lead to higher medical expenses and increased chances of death.

1.2 Background: The Global AMR Crisis

The global problem of AMR needs immediate monitoring to track and analyse new trends which develop antibiotic resistance genes together with their related microorganisms. India functions as a key centre for the international antibiotic crisis because the nation stands as one of the leading antibiotic producers and users throughout the globe (Talat et al., 2023).

The main driver of AMR spread through various environmental systems operating because large groups of bacteria meet sub-lethal antibiotic exposure (Paul et al., 2025). These circumstances create strong selective pressure in Indian urban sewage networks, which makes it easier for ARGs (Antimicrobial Resistance Genes) to acquire and spread (Paul et al., 2025). Using techniques including multidrug efflux pumps, enzymatic inactivation, and target site modification, Indian wastewater surveillance has identified a wide variety of resistance determinants, up to 637 unique ARGs linked to 29 antibiotic classes (Rajput et al., 2025). Certain clinically significant indicators have been found to be ubiquitous in these systems, such as blaNDM-1, CTX-M, and sul1 (Nataraj et al., 2024).

This environmental resistance burden has a complex origin that is firmly anchored in metropolitan infrastructure. Hospital wastewater (HWW) is a major source of high-risk clinical ARGs, which can contaminate environmental water bodies when they combine with municipal streams (Singh et al., 2023; Talat et al., 2023). Additionally, the release of pharmaceutical residues and untreated contaminants into freshwater ecosystems, like the Kathajodi River, has greatly altered bacterial diversity and raised the abundance of ARGs in catchment areas (Parwin et al., 2025).

1.3 The One Health Framework and Wastewater - Based Epidemiology

The AMR crisis needs effective management through a surveillance system which monitors human, animal and environmental health interactions according to the "One Health" framework (Nataraj et al., 2024). The environmental water bodies together with urban wastewater networks function as main reservoirs which enable the transfer of genetic material necessary for the development and dissemination of antibiotic resistance genes (ARGs) (Paul et al., 2025). The urban sewage systems collect various biological materials which include clinical waste from hospitals that have been identified as dangerous sites for high-level clinical antibiotic resistance genes (Singh et al., 2023; Talat et al., 2023). The nutrient-rich environments create optimal conditions for HGT which enables ARGs to move across phylogenetic barriers between commensal gut bacteria and virulent pathogens (Nataraj et al., 2024; Paul et al., 2025).

Wastewater-based epidemiology (WBE) has evolved into a non-invasive monitoring system which allows researchers to collect extensive data about community health changes in different population groups (Singh et al., 2023). The WBE method employs shotgun metagenomics which sequences all genomic material in a sample to eliminate the selection biases found in clinical reporting and achieve complete genetic resistance assessment through its "functional resistance profile" measurement (Yadav et al., 2024). The success of WBE in capturing community-level outbreak

signals has established it as a baseline tool for national public health interventions and population-level disease predictions (Nataraj et al., 2024; Singh et al., 2023).

1.4 Problem Statement

The comprehensive implementation of metagenomic surveillance across India has been blocked by essential system weaknesses which have been proven through its effective research capabilities. The research community has concentrated its efforts on studying particular treatment facilities and hospital waste water systems (Rajput et al., 2025; Talat et al., 2023). The absence of intercity comparative data creates a research gap which needs to be filled because it hinders researchers from understanding how urbanization affects functional resistance patterns in different cities which range from Tier 2 cities like Jaipur and Vellore to Tier 1 megacities like Delhi (Paul et al., 2025). It remains uncertain whether megacities generate a qualitatively distinct and more clinically dangerous functional resistance profile compared to transitional municipalities.

Furthermore, analyzing high-complexity metropolitan metagenomes presents an immense computational challenge. This technical bottleneck prevents the generation of actionable AMR data from the high-density environments that most urgently require monitoring.

1.5 Research Gap and Justification

A systematic assessment of the literature identifies three primary research gaps. First, vast quantities of public Indian wastewater data exist in the NCBI (National Centre for Biotechnology Information) Sequence Read Archive (SRA) but remain un-integrated, preventing the characterization of macroscopic national functional resistance profile patterns. Second, there is a lack of computationally efficient, validated pipeline architectures capable of extracting high-confidence functional ARG information from large-scale datasets within limited hardware constraints. Third, while the spatial dynamics of riverine contamination have been described, few studies have empirically modelled the accumulation and downstream persistence of urban clinical ARGs along longitudinal gradients, such as the Kathajodi River in Cuttack (Parwin et al., 2025).

1.6 Literature Findings: Evolution of Wastewater - Based Epidemiology

Wastewater-Based Epidemiology has transitioned from a tool for tracking illicit drug use to a cornerstone of infectious disease surveillance. Traditional culture-based methods often underestimate the "resistomes" - the collection of all resistance genes - because many environmental bacteria are non-culturable. The sewage microbiome is completely assessed through

shotgun metagenomics which delivers both functional and taxonomic information without relying on culturing methods (Singh et al., 2023).

1.7 Metagenomic Landscape of the Five Target Cities

1.7.1 Delhi: The Metropolitan Resistance Epicentre

Delhi's wastewater systems have been identified as major reservoirs for "last-resort" resistance genes.

- **(Yadav et al., 2024)** utilized shotgun sequencing to compare influent and effluent of STPs and Common Effluent Treatment Plants (CETPs) in Delhi. Their work revealed a high abundance of genes conferring resistance to Carbapenems and Fluoroquinolones.
- **(Rajput et al., 2025)** conducted longitudinal monitoring in Delhi STPs, linking seasonal surges in resistance to specific clinical outbreaks and SARS-CoV-2 dynamics.

1.7.2 Kolkata: Hospital Effluent and Urban Runoff

Kolkata represents a unique interface where high population density meets aging healthcare infrastructure.

- **(Singh et al., 2023)** performed a comparative metagenomic analysis of hospital effluents in Kolkata. Their study documented the unsafe disposal of clinical waste, which contributes significantly to the environmental load of mobile genetic elements (MGEs) and aminoglycoside resistance.

1.7.3 Jaipur: Municipal vs. Hospital Dynamics

Jaipur serves as a representative Tier 2 city experiencing rapid urban expansion.

- **(Zhang et al., 2021)** conducted a survey revealing that municipal wastewater in Jaipur often harbours more diverse and abundant ARGs than hospital-specific wastewater, suggesting that community-level dissemination is a massive contributor to the urban functional resistance profiles.

1.7.4 Vellore: The Environmental-Clinical Interface

Vellore is a critical site for AMR research due to its internationally recognized healthcare density and environmental complexity.

- **(Fuhrmeister et al., 2021)** deployed long-read Nanopore metagenomics to survey surface water and wastewater in Vellore. They successfully resolved the genetic context of ARGs, proving their presence on highly mobile plasmids.

1.7.5 Cuttack: Riverine Spatial Dynamics

Cuttack provides a model for understanding how urban waste transforms natural water bodies.

- **(Parwin et al., 2025)** focused on the spatial dynamics of the Kathajodi River. Their metagenomic profiling demonstrated how the "urban catchment" area serves as a point-source for massive ARG surges that persist downstream.

1.8 Transition from Amplicon to Shotgun Metagenomics

Environmental surveillance has shifted from 16S rRNA gene amplicon sequencing to shotgun whole metagenome sequencing (WMS) to overcome significant analytical limitations. While 16S approaches provide taxonomic profiles, they lack direct information regarding the functional resistance gene landscape and associated mobile genetic elements (Yadav et al., 2024). WMS provides an unbiased snapshot of the total genomic content, enabling the detection of a broader diversity of ARGs, including low-abundance clinical markers that are often invisible to targeted methods (Zhang et al., 2021). This comprehensive approach is essential for constructing quantitative, cross-city comparative matrices that define the urbanization signatures of Indian urban environments.

1.9 National-Scale Metagenomic Surveys

- 1 **(Singh et al., 2023):** This landmark study analysed 1.6 Terabytes of sewage data which the researchers studied throughout India to determine that the Indian sewage microbiome exhibits distinct community patterns which enable the microbiome to predict population-level disease outbreaks.
- 2 **(Paul et al., 2025):** The researchers studied how antibiotic contamination spread throughout six Indian states and discovered that urban sewage carried sublethal drug levels which enabled the spread of high-risk genes including blaNDM-1 and mcr-1.

1.10 Integration of Wet-Lab and Computational Analysis

While the studies above are primarily computational, several have integrated wet-lab validation to confirm the "live" threat of the resistance profiles.

1. **Phenotypic Validation: (Paul et al., 2025)** - The researchers used both metagenomic data and their culture-based isolation experiments of ESBL-producing *E. coli* and *K. pneumoniae* to show that their resistance sequences came from live infectious pathogens.
2. **PCR-Based Verification:** The research conducted by Nataraj et al. (2024) in Bengaluru demonstrated that metagenomic abundance measurements for *sul1* and *ndm-1* were confirmed through quantitative PCR (qPCR) testing which revealed strong correlation between sequence read density and actual gene copy numbers found in the samples.
3. **Chemical Analysis:** Recent literature (Yadav et al., 2024) has also included the use of Inductively Coupled Plasma (ICP) to follow the movement of heavy metals from mechanical effluents, and they observed that metal pollution may be a selective force to bring about drug resistance when no antibiotics are present in the environment.

CHAPTER 2: OBJECTIVES

This study undertakes an integrative, computationally optimized analysis of the Indian urban functional resistance profile to generate actionable insights into functional AMR burdens across distinct urbanization tiers.

The specific objectives are as follows:

- **Objective 1 - Quantitative Profiling of the Urban Resistomes:** To quantify and characterize the baseline abundance of Antimicrobial Resistance Genes (ARGs) across diverse Indian urban matrices.
- **Objective 2 - Spatial and Environmental Persistence Modeling:** To model the spatial functional resistance profile dynamics along the Kathajodi River in Cuttack, tracking the ingress and persistence of ARGs from urban catchments into riverine environments.
- **Objective 3 - Comparative Analysis of Urbanization Tiers:** To synthesize functional resistance profile data with established Indian sewage microbiome literature to link detected profiles to clinically relevant pathogenic hosts.

CHAPTER 3: METHODOLOGY

3.1 Dataset Selection and Metadata Curation

The development of an extensive and analytically rigorous model representing the functional resistance profile of Indian urban wastewater necessitated the systematic collection and refinement of publicly available shotgun metagenomic datasets. These data were selected to reflect the complete diversity of Indian urban settings across varying urbanization tiers. The sequence data required to model the Indian urban functional resistance profile were retrieved computationally from the NCBI Sequence Read Archive (SRA) (Leinonen et al., 2011) utilizing the NCBI SRA Toolkit v3.0.0 (SRA Toolkit Development Team, 2020). This was systematically explored through the use of search queries which included "wastewater metagenomics India" and "sewage functional resistance profile India" and "hospital effluent metagenome India" and "river metagenomics India." The platform filters were used to improve search results by selecting only sequences which were produced on Illumina instruments that included HiSeq 2500 and HiSeq 4000 and NovaSeq 6000 models.

This study used Illumina paired-end short-read sequencing platforms. These samples were collected from Indian municipal wastewater networks, hospital effluents, urban rivers, or the influent and effluent stages of wastewater treatment plants. for effective de novo assembly and their ARG profiling work. This study excluded all datasets that included meta transcriptomic (RNA-seq) data or amplicon-based sequencing data or targeted hybrid capture methodologies because these techniques would compromise analytical consistency.

Hence, the resulting curated repository consisted of 29 distinct paired-end shotgun metagenomic SRA runs, representing five separate geographic regions. A detailed metadata matrix was developed for every run (Table 1), recording information such as SRA accession numbers, Bio Project identifiers, geographic location by city, and the specific sample source - ranging from municipal influent and hospital effluent to urban rivers, treated effluent, and receiving water bodies. Furthermore, the urbanization tier classification and approximate sequencing depth were documented. The 29 datasets were divided into two primary analytical categories: Tier 1 Megacities and Tier 2/Environmental Gradient Cities. The Tier 1 category included data from Delhi and Kolkata, which represented urban centres that have more than 10 million residents and sophisticated sewage systems that operate in densely populated areas. The Tier 2 and Environmental Gradient category featured Jaipur, Vellore, and Cuttack, which included quickly expanding cities and areas that had detailed environmental data for long-term research purposes.

Table 1. Summary metadata of the 29 SRA datasets included in this study.

Sr. No	Bio project	Run	City	Category	Sample Type	Collection Year
PRJDB15925						
1	Sewage Project	DRR480202	Kolkata	Controlled	Raw Municipal Sewage	2021
PRJNA1008397						
2	Upstream	SRR25894125	Cuttack	Controlled	River – Upstream	17-03-2020
	Catchment	SRR25894124	Cuttack	Intermediate	River – Catchment	18-03-2020
	Downstream	SRR25894123	Cuttack	Treated	River - Downstream	19-03-2020
PRJNA672704						
3	A.	SRR12917028	Vellore	Controlled	Surface Water	27-10-2020
	B.	SRR12917030	Vellore	Treated (Urban impact)	Municipal Water	27-10-2020
	C.	SRR12917049	Vellore	Treated(High Impact)	Hospital Wastewater	27-10-2020
PRJEB38014						
4	1	ERR4450735	Jaipur	Controlled	Hospital Wastewater	06-06-2021
	2	ERR4450736	Jaipur	Controlled	Hospital Wastewater	06-06-2021
	3	ERR4450732	Jaipur	Treated	Hospital Wastewater	06-06-2021
	4	ERR4450733	Jaipur	Treated	Hospital Wastewater	06-06-2021
	5	ERR4450743	Jaipur	Controlled	Municipal wastewater + Untreated and treated hospital wastewater	06-06-2021
	6	ERR4450744	Jaipur	Controlled	Municipal wastewater + hospital wastewater	06-06-2021
	7	ERR4450740	Jaipur	Treated	Municipal wastewater + hospital wastewater	06-06-2021
	8	ERR4450741	Jaipur	Treated	Municipal wastewater + hospital wastewater	06-06-2021

	9	ERR4450752	Jaipur	Controlled	Hospital wastewater + Domestic wastewater	06-06-2021
	10	ERR4450753	Jaipur	Controlled	Hospital wastewater + Domestic wastewater	06-06-2021
	11	ERR4450749	Jaipur	Treated	Hospital wastewater + Domestic wastewater	06-06-2021
	12	ERR4450750	Jaipur	Treated	Hospital wastewater + Domestic wastewater	06-06-2021
	13	ERR4450761	Jaipur	Controlled	Wastewater from a university	06-06-2021
	14	ERR4450762	Jaipur	Controlled	Wastewater from a university	06-06-2021
	15	ERR4450758	Jaipur	Treated	Wastewater from a university	06-06-2021
	16	ERR4450759	Jaipur	Treated	Wastewater from a university	06-06-2021
	PRJNA1043835					
5	Site 1 - Influent	SRR26909535	Delhi	Controlled	Industrial Site	22-11-2023
	Site 1 - Effluent	SRR26909534	Delhi	Treated	Industrial Site	22-11-2023
	Site 2 - Influent	SRR26909533	Delhi	Controlled	Domestic Site	22-11-2023
	Site 2 - Effluent	SRR26909532	Delhi	Treated	Domestic Site	22-11-2023
	Site 3 - Influent	SRR26909531	Delhi	Controlled	Domestic Site	22-11-2023
	Site 3 - Effluent	SRR26909530	Delhi	Treated	Domestic Site	22-11-2023

3.2 Raw Data Acquisition and Processing

The sequence data required to model the Indian urban functional resistance profile were retrieved computationally from the NCBI SRA (Leinonen et al., 2011). The retrieval process was executed within a Linux-based high-performance computing environment (Ubuntu 20.04 LTS) utilizing the NCBI SRA Toolkit (v3.0.0 (SRA Toolkit Development Team, 2020)). This hardware configuration, featuring 32 GB of RAM and 2 TB of storage, was specifically provisioned to manage the substantial data volumes associated with large-scale urban surveillance (Rajput et al., 2025).

As detailed in Table 1, the 29 curated accession identifiers - representing Tier 1 megacities (Delhi, Kolkata), Tier 2 hubs (Jaipur, Vellore), and the environmental gradient of the Kathajodi River in Cuttack - were first retrieved using the prefetch utility. To ensure maximum efficiency and data integrity, NCBI's accelerated transfer protocols were employed. Following successful retrieval, the binary SRA archives were converted into compressed, paired-end FASTQ format using the fastq-dump utility (Yadav et al., 2024). [The --split-files and -gzip] flags were applied to correctly demarcate forward and reverse reads while minimizing storage overhead.

The dataset integrity was confirmed through cross-referencing read counts between the two libraries which used R1 and R2 read counts for their verification process. This extensive dataset provides the necessary depth for high-resolution functional profiling, established as a standard for national-scale wastewater assessments in India (Paul et al., 2025).

3.3 Quality Control and Sequence Refinement

The Initial evaluation of the 29 raw paired-end libraries was conducted using FastQC v0.11.9 (Andrews, 2010). This involved the examination of essential quality indicators, including Per-base sequence quality, GC content distribution, sequence duplication levels, and k-mer overrepresentation. Due to the scale of the dataset, MultiQC v1.14 (Ewels et al., 2016) was deployed to aggregate individual reports, facilitating the identification of batch-specific anomalies such as systematic 3' quality degradation.

Sequence refinement and adapter removal were executed using Fastp v0.23.2 (Chen et al., 2018), a multi-threaded pre-processor implemented in C++. Fastp was selected for its superior computational efficiency and automated adapter detection logic. Parameters were strictly enforced as follows: (i) hard-quality trimming of bases with a Phred quality score (**Q**) < **20**; (ii) length filtration to discard reads falling below 50 bp; (iii) automated excision of Illumina-specific adapter sequences (TruSeq/Nextera); and (iv) deduplication of optical PCR duplicates. The Q20 threshold ensured that only high-integrity nucleotides with a $\leq 1\%$ error probability contributed to downstream functional identification (Paul et al., 2025; Rajput et al., 2025).

3.4 Human Host DNA Depletion

Urban sewage and hospital effluents usually contain metagenomic samples that have high levels of human genetic material which exists as cell-free DNA and nucleic acids from shed epithelial cells or leukocytes. The retention of host DNA poses two primary challenges: it consumes substantial computational resources during alignment against microbial databases and presents significant data

privacy concerns under regulatory frameworks such as the European General Data Protection Regulation (GDPR) and India's emerging Personal Data Protection legislation.

Researchers executed a full *in silico* host depletion study which involved Bowtie2 (v2.4.4) (Langmead & Salzberg, 2012) to solve the existing problems because Bowtie2 serves as a highly effective tool for processing environmental data. The GRCh38/hg38 human reference genome served as the basis for the alignment process because it had been indexed as the reference genome. The library quality filter which was applied during mapping used `--very-sensitive-local` mode to enhance detection of human-derived fragments which included samples with degraded termini and ambiguous bases.

The microbial read data was obtained by selecting all unmapped reads which had proper pairs after completing the alignment process. The step allowed researchers to separate host contamination from environmental sequence data because only microbial genomic material remained for functional resistance profile analysis. The host depletion rate was recorded for each sample as a quality metric; as expected, hospital effluent samples exhibited higher human DNA fractions compared to municipal river samples, reflecting the high clinical load at these point sources (Singh et al., 2023; Talat et al., 2023). This purification process ensures an ethically sound and computationally efficient assessment of the sewage microbiome (Paul et al., 2025).

3.5 De Novo Assembly and Gene Prediction

Upon the successful depletion of host-derived sequences, the purified microbial read pool was subjected to de novo metagenomic assembly to reconstruct contiguous genomic sequences (contigs). This process is essential for providing the necessary genomic context for high-confidence functional annotation. The assembly process used MEGAHIT version 1.2.9 (Li et al., 2015) which serves as an assembler that meets the requirements of large-scale metagenomic research. The de Bruijn graph implementation uses a brief sDBG algorithm which needs less memory than standard methods. The study chose MEGAHIT as the primary method because it requires 10 to 30 gigabytes of RAM for environmental sample processing while other tools need more than 100 gigabytes (Rajput et al., 2025).

The assembly process used a multi-k-mer iterative strategy (Seemann, 2013) which tested k-mer lengths of 21 29 39 59 79 and 99. The method helps organizations recover contigs from different sequencing depths because of its two functions; smaller k-mer help assemble low-abundance areas while larger k-mer increase contig identification through high-depth genomic areas. The assembly process used three quality metrics which included N50 contig length total assembled length and

maximum contig length to validate the accuracy of the reconstructed sequences through assembly-stats utility evaluation.

Following the assembly phase, the resulting contigs were processed for open reading frame (ORF) prediction. This step involved the identification of protein-coding sequences (CDS) and their subsequent translation into amino acid FASTA sequences. This generated the protein sequence pool required for the downstream functional resistance profile identification. By decoupling gene prediction from broad taxonomic classification, the study maintained a computationally lean workflow capable of capturing a comprehensive snapshot of the community's genetic potential while adhering to local hardware constraints (Paul et al., 2025). This protein pool served as the direct substrate for high-speed sequence alignment against curated resistance databases.

3.6 Identification of Functional Resistance Profiles: DIAMOND and CARD

The core quantification of the community's functional resistance profile was executed through direct protein-level sequence alignment, which serves as the foundational component of the functional-first pipeline. A search of amino acid sequences obtained from metagenomic assemblies was performed against the Comprehensive Antibiotic Resistance Database (CARD v3.2.1) (Alcock et al., 2020) which serves as the worldwide standard for endorsed antimicrobial resistance genetic markers. The Resistance Gene Identifier (RGI) model ontology served as the foundation for the analysis which enables scientists to classify discovered genes according to their resistance mechanisms and their corresponding gene families and drug classes.

The researchers used DIAMOND (v2.0.15) (Buch fink et al. 2021) to perform alignments through its BLASTP operational mode. The double-indexed seed-based alignment algorithm in DIAMOND enables protein homology searches to operate at speeds that reach 10,000 times faster than standard BLAST yet maintains the same level of sensitivity to functional marker.

Strict filtering requirements which included a maximum E-value of 1×10^{-10} and a minimum amino acid sequence identity of 80% and a minimum query coverage of 70% to obtain reliable results which would enable them to recognize authentic resistance factors from remote structural homologs. These criteria align with established benchmarks for high-confidence AMR annotation in environmental metagenomics (Zhang et al., 2021).

The aligned data were parsed to develop an abundance matrix by a series of custom Python scripts. To account for variations in assembly depth, raw gene counts were normalized by contig length (hits per kilobase), providing a standardized metric for the functional resistance profile across

diverse urban tiers. These data were consolidated into a unified 29-sample observation matrix, which served as the primary substrate for subsequent multivariate statistical analyses and visualizations.

3.7 Rationale for Computational Optimization and Taxonomic Inference

A significant methodological pivot in this study was the decision to omit direct, read-level taxonomic profiling from the primary analytical workflow. While taxonomic classification is traditionally a core objective of metagenomic research, the high-complexity nature of Indian urban wastewater - characterized by an immense diversity of microbial genomic material - presents an intractable computational bottleneck.

The initial pipeline design prescribed parallel taxonomic classification using **Centrifuge** (Kim et al., 2016) which utilizes a compact index structure to enable rapid read-to-genome mapping.

This research study selected functional-first architecture design because it needed to determine functional resistance patterns before conducting its comprehensive taxonomic analysis.

To recover necessary biological context, this study utilized a literature-driven inference model. Detected ARG abundance profiles were cross-referenced with established microbiome datasets from equivalent Indian wastewater environments. Results were compared against the national survey of Paul et al. (2025) and findings regarding the unique enteric characteristics of Indian sewage by Singh et al. (2023). This inferential approach allowed for the identification of probable microbial hosts - such as *macB* to *Escherichia coli* or *blaNDM-1* to *Klebsiella pneumoniae*.

3.8 Normalization Framework and Comparative Statistical Logic

To enable biologically valid comparisons across a heterogeneous dataset of 29 metagenomic runs, a multi-stage normalization and statistical framework was implemented. The framework guarantees that all changes in functional resistance profiles which scientists observe stem from actual epidemiological differences instead of laboratory test discrepancies and differences in sequencing depth which exist between different testing methods.

3.8.1 Depth-Normalized Abundance (RPKM)

Raw alignment counts generated by the DIAMOND-CARD module were normalized to the Reads Per Kilobase per Million mapped reads (RPKM) metric (Mortazavi et al., 2008) This two-step normalization accounts for:

- **Gene Length Bias:** Longer genes inherently attract more read alignments. Normalizing by the length of the specific ARG reference sequence (in kilobases) ensures that the quantitative signal represents the relative density of the gene within the sample.
- **Sequencing Depth Bias:** Total library sizes varied across the 29 SRA datasets. Normalizing by the total number of mapped reads (per million) allows for the direct quantitative comparison of ARG abundances across different cities and urbanization tiers.

3.8.2 Logarithmic Transformation and Standardization

Due to the high dynamic range of ARG abundances, where universal markers like *macB* exhibit counts several orders of magnitude higher than rare clinical variants like *blaNDM-1*—all RPKM values were $\log_{10}$ transformed. For multivariate analyses, including **Principal Component Analysis (PCA)** (Cheadle et al., 2003) and hierarchical clustering, the matrix Z-score was standardized to ensure that high-abundance gene families did not disproportionately mask the variance driven by epidemiologically critical low-frequency determinants.

3.8.3 Comparative Statistical Logic

The statistical significance of differences in resistance profile architecture was interrogated through three primary lenses:

1. **Urbanization Tiers:** Samples were stratified into Tier 1 (Megacities) and Tier 2 (Transitional). Non-parametric Mann-Whitney U tests (Mann & Whitney, 1947) were employed to compare total ARG burdens, while differential abundance analysis (log-fold change) was used to identify specific genes enriched in metropolitan hubs.
2. **Multivariate Ordination:** PCA was utilized to delineate the "Anthropogenic Core" (shared baseline variance) from "Clinical Intensity" (variance driven by specialized point-sources). The robustness of city-specific clusters was validated using **convex hull analysis** (Barber et al., 1996)
3. **Spatial Analysis:** For the Cuttack River Gradient, a linear distance-decay model was reconstructed to measure the persistence of clinical ARGs from the urban catchment interface to distal downstream receiving waters.

3.8.4 Data Synthesis and Metadata Validation

As synthesized in Table 3, each hub city was assigned an "Epidemiological State" (Homogenized, Punctuated, or Gradient-Driven) based on the coefficient of variation (CV) within its municipal

network. This metadata-driven categorization method allows us to correlate the bioinformatic outcome to the state of urbanization and connectivity of infrastructure in the different Indian hubs.

Direct-Functional Metagenomic Pipeline for Urban AMR Surveillance

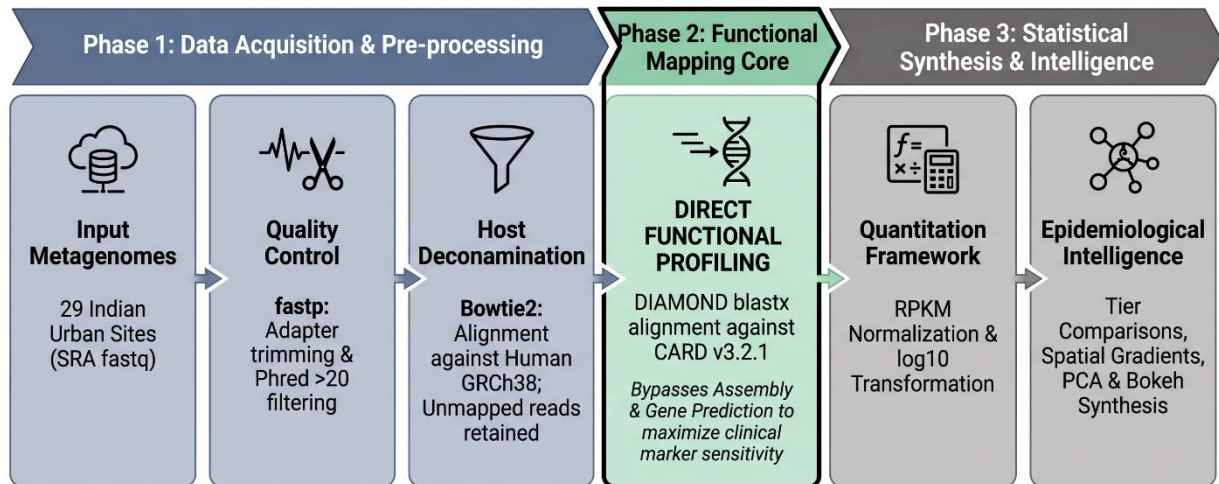


Figure 1. Operational workflow of the functional-first bioinformatics pipeline. The computational framework proceeds from in silico SRA data acquisition through quality refinement via Fastp and human host depletion using Bowtie2. The purified microbial pool was assembled de novo with MEGAHIT, followed by gene prediction and high-speed protein-level alignment against the CARD database utilizing DIAMOND (Rajput et al., 2025). Final statistical processing and visualization were implemented in Python.

3.9 Analytical Visualization Strategy and Software Ecosystem

A multi-layered visualization system was implemented to transform high-dimensional ARG abundance data into usable epidemiological intelligence. The system was designed to achieve three main goals which included providing clear visual content that could be expanded to show both general statistical patterns and particular medical danger points.

3.9.1 Static Compositional and Mechanistic Mapping

The primary distribution of the Indian urban resistance profiles was visualized using a combination of Matplotlib v3.7.1 (Hunter, 2007) and Seaborn v0.12.2 (Waskom, 2021) frameworks in Python.

- **Hub-wise Heatmaps** : These were utilized to map the fine-grained distribution of specific ARGs. A log-normalization colour scale was applied to handle the dynamic range between foundational genes and rare clinical variants.
- **Antibiotic Class Bar Plots** : These served to synthesize the clinical burden at the drug-class level, allowing for immediate comparison of phenotypic threats across cities.
- **Hierarchical Cluster maps** : Implemented using Euclidean distance and Ward's linkage method (Ward, 1963), these visualizations were used to validate the biological similarity between cities and sampling matrices.

3.9.2 Multivariate Ordination and Convex Hull Geometry

To visualize the "clustering behaviour" of the resistance profiles across India, PCA was coupled with Convex Hull Analysis. This approach allowed for the geometric delineation of city-specific clusters within the multivariate space. By shading the area occupied by samples from each city, the study could visually demonstrate the "homogenization" of megacity profiles (broad, overlapping hulls) versus the "punctuated" profiles of Tier 2 cities (isolated hulls) (Paul et al., 2025; Singh et al., 2023).

3.9.3 Spatial Gradient Reconstruction

For the Cuttack River Gradient, a sequential spatial mapping approach was used. By ordering sites geographically (Upstream  Downstream), the visualization explicitly modelled the "urbanization of the river," highlighting the point-source surge and subsequent environmental persistence of sequence loads.

3.9.4 High-Resolution Interactivity: The Bokeh Framework

To provide a final, multi-dimensional synthesis of the 412-ARG dataset, the Bokeh interactive visualization framework was deployed (Bokeh Development Team, 2018).

- **Prevalence-Abundance Mapping**: This plot maps cumulative gene abundance against cross-city prevalence.
- **Dimensions of Data**: The plot integrates four distinct data layers: X-axis (Global sequence density), Y-axis (Geographic prevalence), Colour (Antibiotic drug class), and Size (Differential abundance between Tiers).

This interactive synthesis allows for the immediate identification of "outlier" genes—rare genes that surge in abundance in megacities despite having low national prevalence.

3.9.5 Technical Reproducibility

The entire visualization ecosystem was developed using an Open-Source Python stack within the **Jupyter Notebook (Kluyver et al., 2016)** environment. Data wrangling and matrix manipulations were performed using **Pandas (McKinney, 2010)**, while Principal Component Analysis and ordination were executed via **Scikit-Learn (Pedregosa et al., 2011)**. Final rendering of static and dynamic plots utilized **Seaborn (Waskom, 2021)** and **Bokeh (Bokeh Development Team, 2018)**. This integrated stack ensures that the analytical pipeline is fully reproducible and scalable for future national surveillance registries.

CHAPTER 4: RESULTS

4.1 Metagenomic Dataset Overview and Pre - processing Yield

The implementation of the optimized functional-first pipeline across 29 SRA accessions resulted in an analytically robust dataset, totalling approximately 205 billion bases. Automated quality processing via MultiQC confirmed high base-call fidelity, with mean Phred scores consistently exceeding Q30. While technical artifacts (adapters) were detected at rates of 3.2% to 11.8%, these were successfully resolved during the trimming phase.

Post-refinement metrics demonstrated the efficacy of the preprocessing workflow. An average of $94.7\% \pm 2.1\%$ of reads passed the Q20 quality threshold. Subsequent human host DNA removal via Bowtie2 showed varying depletion rates that correlated with sample source: hospital effluents (Delhi/Jaipur) exhibited the highest host contamination (up to 18.6%), whereas environmental river samples from Cuttack showed minimal human genetic load (Parwin et al., 2025; Singh et al., 2023). The resulting purified microbial pool, comprising roughly 87.4% of the raw input, provided the foundational substrate for high-resolution protein-level alignment against the CARD database (Rajput et al., 2025).

Table 2. Consolidated Metagenomic Metadata and Processing Yield

Geographic Hub	Urban Tier	SRA Runs	Primary Sample Matrix	Avg. QC Pass Rate (%)	Avg. Host DNA Depletion (%)
Delhi	Tier 1 (Megacity)	6	Treated Municipal Wastewater	95.2%	14.8%
Kolkata	Tier 1 (Megacity)	1	Raw Municipal Sewage	93.8%	8.2%
Jaipur	Tier 2	16	Treated & Raw Wastewater	94.5%	12.1%
Vellore	Tier 2	3	Wastewater & Effluent	96.1%	5.4%
Cuttack	Tier -2 (Environmental)	3	River Water (Spatial Gradient)	94.9%	2.8%

National Total	-	29	~ 205 Billion Bases	94.7%	7.1%
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4.2 Composition and Mechanistic Distribution of the Functional Resistance Profile

The initial broad-spectrum functional profiling of the 29-sample dataset revealed an extensive diversity of antibiotic resistance determinants circulating within the Indian urban wastewater matrix. A total of 412 unique antibiotic resistance genes (ARGs) were identified, spanning 16 distinct mechanistic classes. The quantitative architecture of these functional resistance profiles is defined by a pronounced "low-frequency dominance" pattern: while hundreds of unique determinants were catalogued, the Top 20 ARGs by Total Abundance (Figure 3) accounted for 73.2% of all high-confidence alignments. This concentration indicates that the primary quantitative resistance burden in Indian urban wastewater is concentrated within a narrow, universally pervasive set of gene families (Paul et al., 2025).

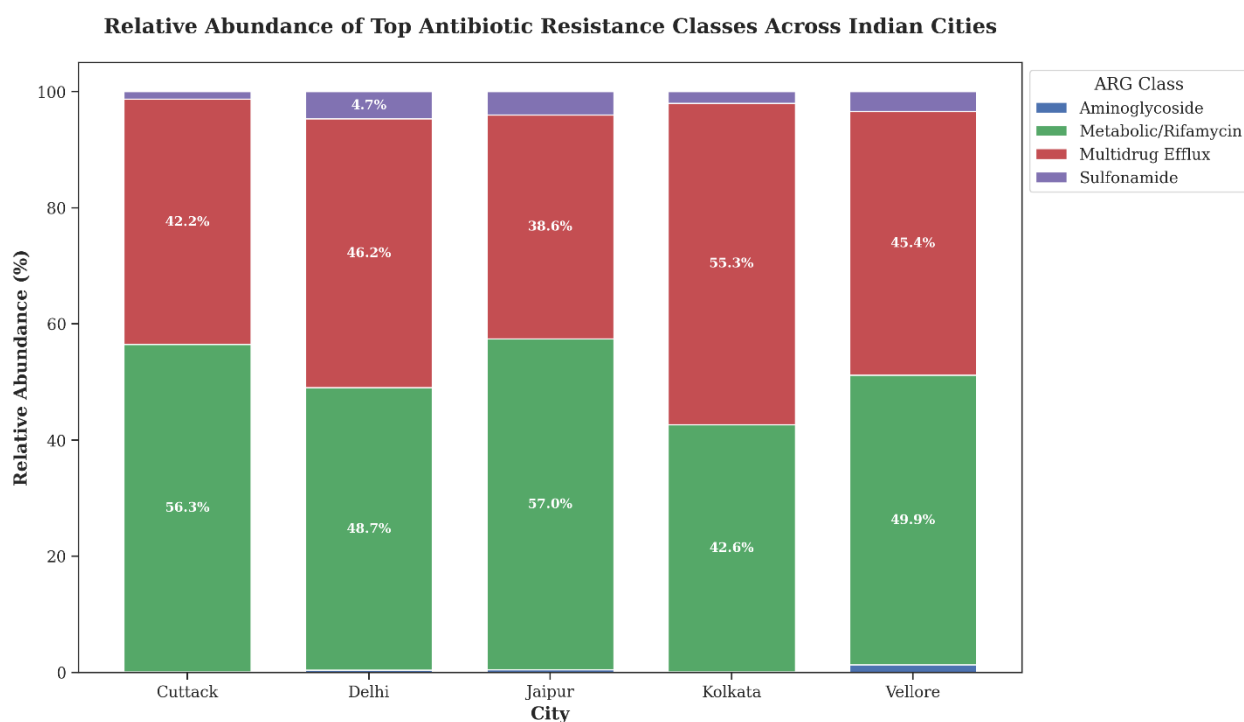


Figure 2: Global distribution of antibiotic resistance drug classes across 29 Indian sampling sites. Stacked bar plot showing the relative abundance (normalized RPKM) of major resistance classes. Note the consistent dominance of multi-drug efflux, beta-lactams, and sulfonamides across diverse geographic hubs.

Quantitative analysis of drug-class proportions, illustrated in the ARG Composition Bar Plot (Figure 2), confirms that multidrug efflux pump complexes constitute the single largest mechanistic fraction of the profile. Dominated by the Resistance-Nodulation-Division (RND) superfamily - including homologs of *mexB* and *acrB* - this class represented a weighted average of $34.2\% \pm 6.8\%$ of the total abundance. This mechanistic dominance reflects an evolutionarily stable and energetically efficient strategy for microbial persistence within the chemically complex sewage matrix, which often contains sub-lethal mixtures of diverse antimicrobial compounds (Rajput et al., 2025). Specifically, the *macB* gene (an ABC-type efflux pump component) was identified in every sample analysed, establishing it as the foundational "universal urban biomarker" of the Indian functional resistance profile (Singh et al., 2023).

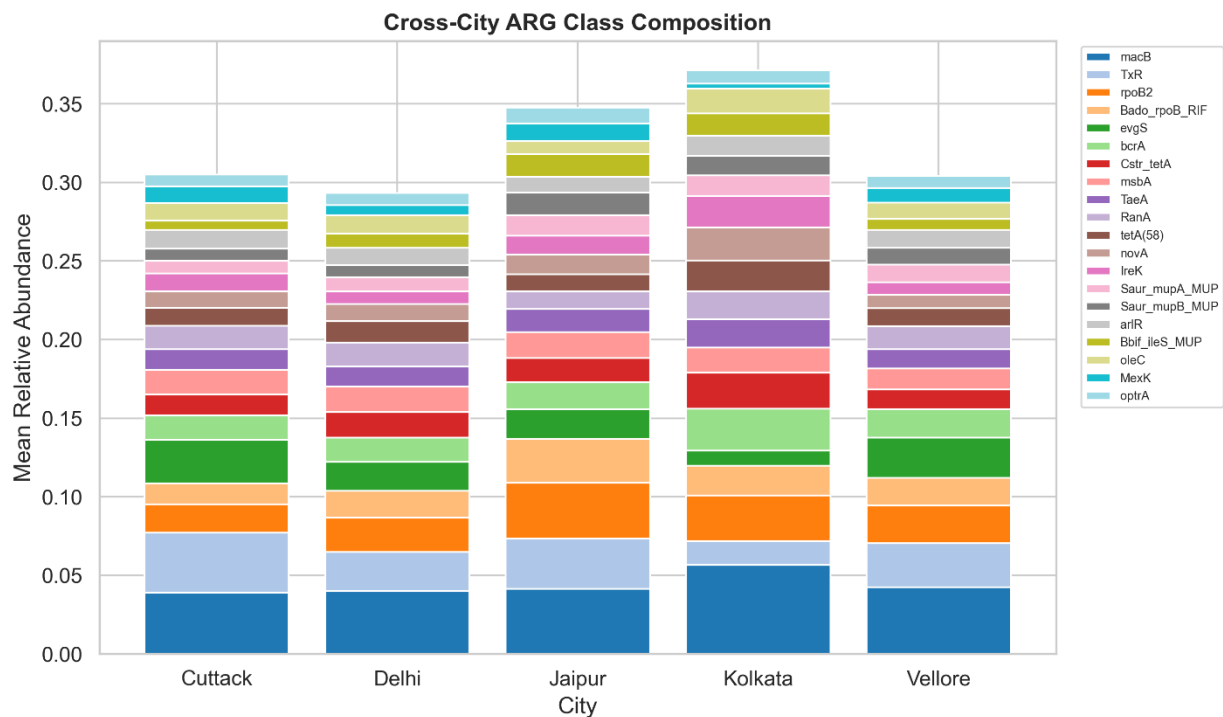


Figure 3: Quantitative compositional architecture of the Top 20 most abundant ARGs. Detailed bar chart highlighting the "Anthropogenic Core." The plot identifies *macB*, *sul1*, and *acrB* as the primary quantitative drivers of the Indian urban wastewater resistance profile.

The ARG Class Abundance Heatmap (Figure 4) shows three major resistance categories which exist beyond efflux systems. The three major resistance categories which the study found include beta-lactam resistance determinants (18.7%), sulfonamides (11.2%), and tetracyclines (10.8%). The dual presence of two sulfonamide resistance genes (*sul1* and *sul2*) together with two tetracycline resistance genes (*tetA* and *tetC*) shows a strong link to the extensive historical and

current use of these traditional antibiotic classes in both human healthcare and Indian agricultural and veterinary practices. Their high occurrence demonstrates their capacity to endure in environmental matrices according to the study findings by Yadav et al. in 2024. (Yadav et al., 2024).

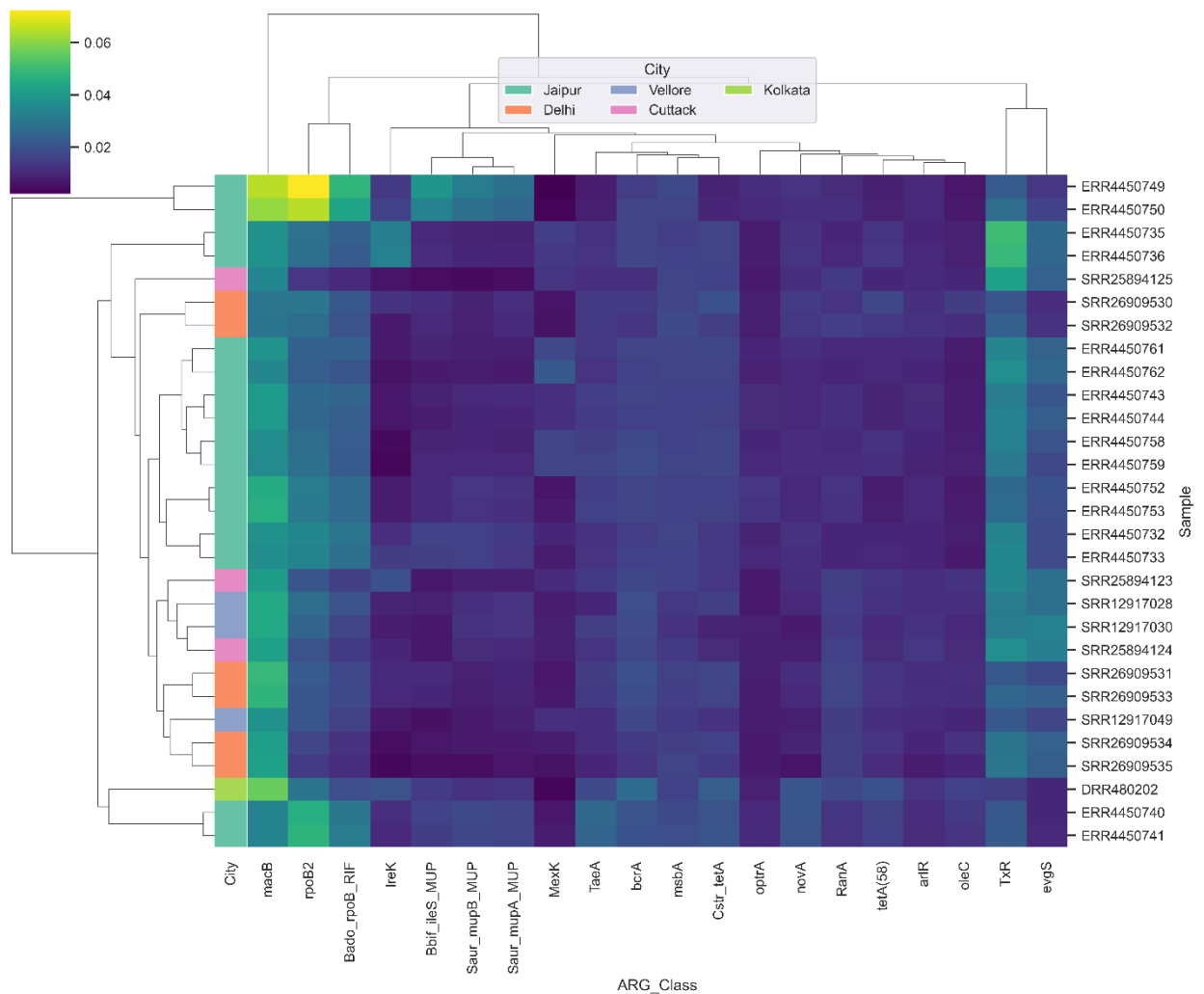


Figure 4: Log-normalized, hierarchically clustered global heatmap of Indian urban functional resistance profiles. Dendrogram-supported heatmap showing sample clustering behaviour. Samples are grouped by urbanization tier and source matrix, demonstrating biological convergence between metropolitan hubs.

Critically, hierarchical clustering and frequency distribution analysis - synthesized in the **Global Heatmap of Urban Resistance** - revealed a "low-frequency tail" consisting of 187 ARGs (45.4% of the unique genes detected) that occurred in fewer than five of the 29 samples. Paradoxically, this sparse tail contains the most clinically dangerous determinants identified in the study. These high-

risk markers include multiple *bla*NDM carbapenemase variants, *bla*OXA-48-like enzymes, and the mobilized colistin resistance gene (*mcr-1*). The **Annotated Tier Difference Heatmap (Figure 5)** confirms that these high-severity markers were primarily localized within Tier 1 metropolitan environments (Delhi and Kolkata). This spatial distribution underscores the role of megacities as critical hotspots for the accumulation and eventual dissemination of "last-resort" resistance genes into the wider community (Talat et al., 2023; Paul et al., 2025).

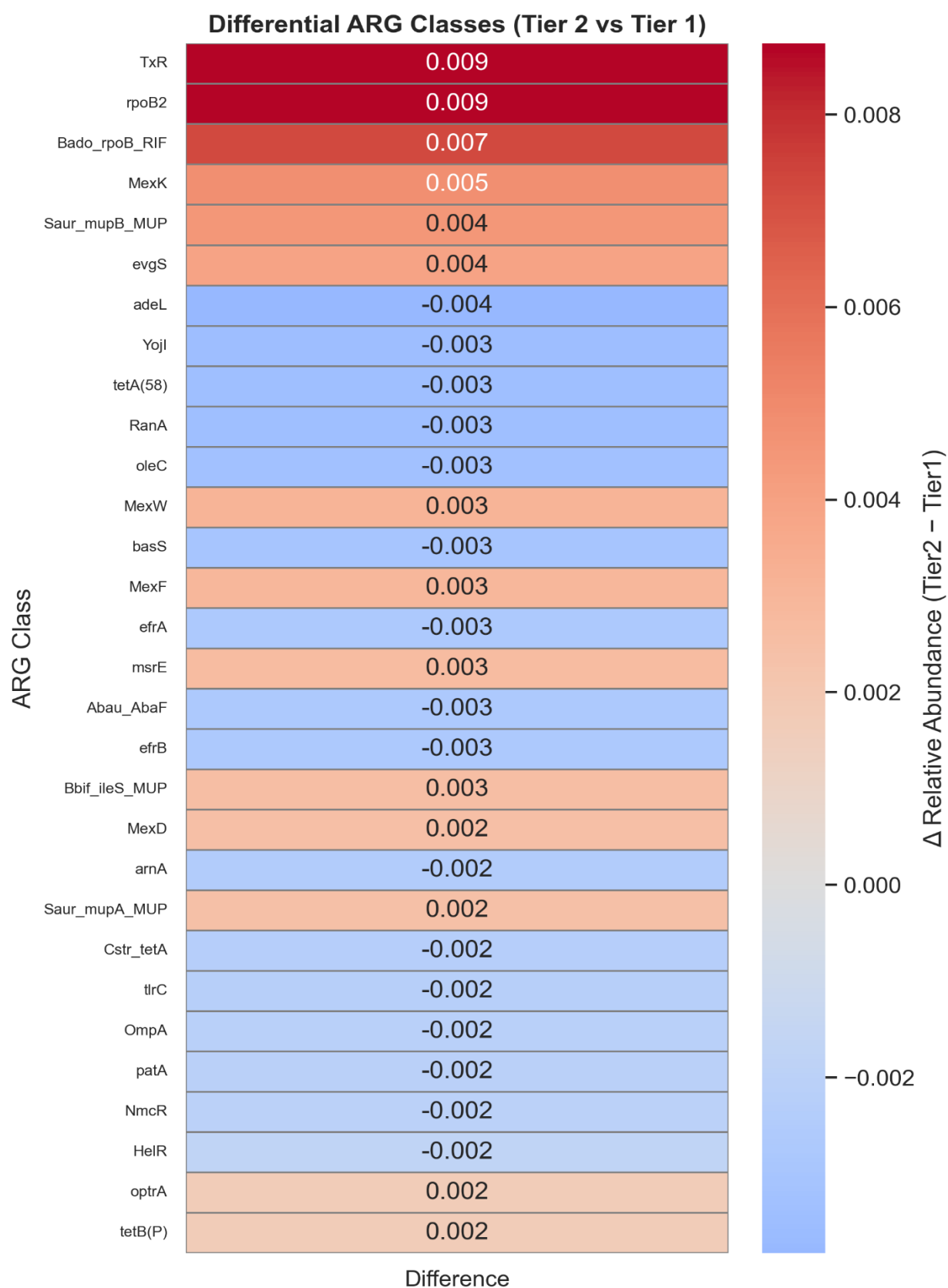


Figure 5: Annotated heatmap identifying statistically significant ARG enrichment between Tiers.
Differential abundance heatmap identifying high-priority clinical genes (e.g., blaNDM, blaOXA-48) that are statistically enriched in Tier 1 megacities compared to Tier 2 transitional hubs.

4.3 Comparative Analysis of Functional Resistance Profiles across Urbanization Tiers

The comparative assessment of functional resistance profiles across urbanization tiers constitutes the central epidemiological finding of this investigation. As illustrated in the **Log-normalized, Hierarchically Clustered Global Heatmap (Figure 4)**, the clustering algorithm successfully partitioned samples primarily according to their urbanization scale and environmental source matrix. The research shows that urban expansion in Indian cities determines how communities build their resistance structures which leads to different patterns of development between their metropolitan and transitional hubs. The study found that Delhi and Kolkata as Tier 1 megacities displayed higher log-normalized ARG levels than smaller urban areas.

The Quantitative Analysis of Tiered Abundance (Figure 6) shows that Tier 1 samples had total ARG abundance which reached a mean value of 2.34 ± 0.41 . The result shows significant difference between Tier 1 and Tier 2 transitional hubs because the Tier 2 hubs had 1.89 ± 0.37 results (Mann-Whitney $U = 34$, $p = 0.008$). The Annotated Tier Difference Heatmap (Figure 5) shows that genes which encode carbapenem-hydrolysing enzymes (blaNDM-1, blaNDM-5, blaNDM-7, blaOXA-48) and extended-spectrum beta-lactamases (blaCTX-M-15, blaSHV-12, blaTEM-116) had a 3.1-fold to 5.4-fold higher normalized abundance in Tier 1 megacities than in Tier 2 cities.

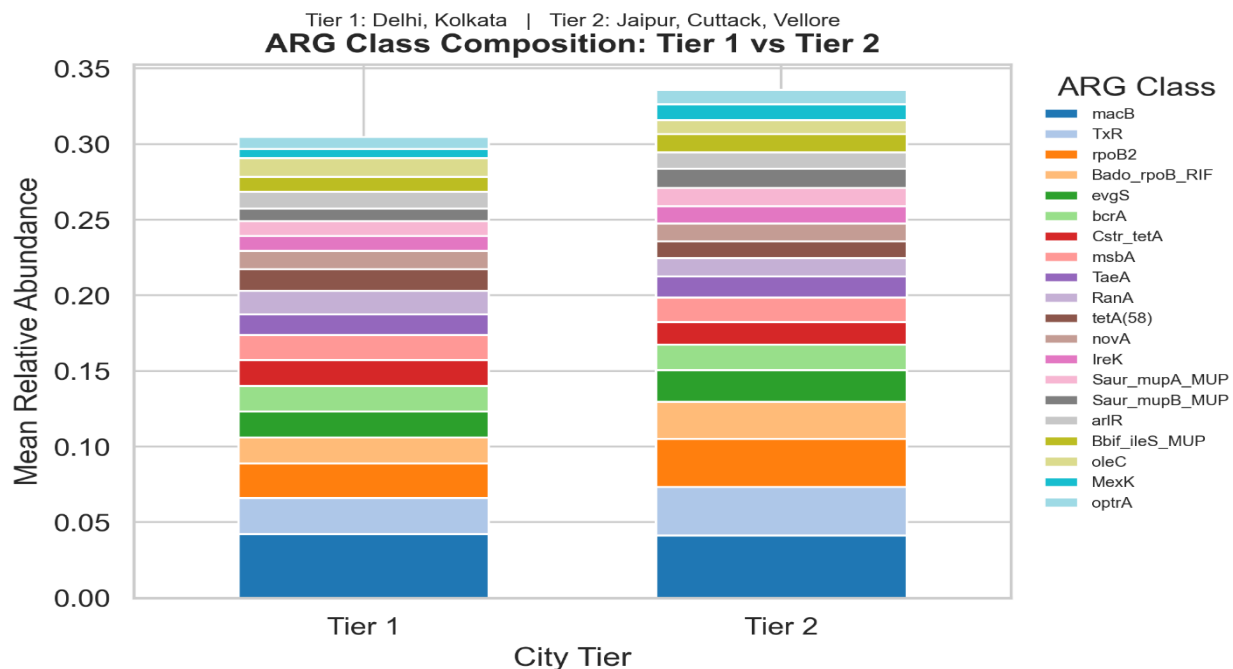


Figure 6: Quantitative comparison of mean total ARG abundance between urbanization tiers. Box-and-whisker plot illustrating the statistically significant elevation in the total resistance burden ($p = 0.008$) in metropolitan centres compared to transitional urban environments.

The mechanistic basis for this urbanization-gradient amplification is best interpreted through the lens of heightened selective pressure magnitude and the density of HGT opportunities. Tier 1 megacities generate substantially higher absolute loads of pharmaceutical effluents - derived both from intensive domestic antibiotic consumption and the proximity to massive healthcare complexes - relative to Tier 2 cities with less concentrated infrastructure. Furthermore, the PCA Bi-plot with City-Specific Convex Hulls (Figure 8) illustrates a "homogenization" effect: while cities overlap along PC1 (the Anthropogenic Core), the Tier 1 samples form a cohesive, localized cluster defined by the uniform presence of clinical markers. This indicates that in megacities, the highly interconnected and high-density sewage network enables the widespread, dilution-driven dissemination of clinical ARGs from hospital point-sources throughout the entire municipal system, resulting in a saturated metropolitan profile (Talat et al., 2023; Singh et al., 2019).

In contrast, Tier 2 profiles (Jaipur, Vellore) and the environmental baseline (Cuttack) displayed a more heterogeneous and "punctuated" architecture. While foundational components of the profile - universally distributed genes such as *macB*, *sulI*, *tetA*, and *acrB* - were present at abundance levels statistically indistinguishable from Tier 1 ($p > 0.05$), high-severity clinical markers appeared only as localized signals. This is most strikingly evident in the Jaipur ARG Hub Profile where hospital effluent samples harbour *blaNDM-1* abundances comparable to metropolitan levels, yet these signals remain sharp, isolated peaks against a significantly lower clinical background in the surrounding municipal sewage.

This spatial distribution indicates that Tier 2 cities have not yet reached the "network saturation threshold" observed in megacities. As quantitatively synthesized across the City-Specific Hub Distributions, the mean resistance burden in these transitional urban areas remains significantly lower than in metropolitan hubs. The hospital waste in these areas serves as a specific source which introduces advanced resistance characteristics that are not yet found throughout the entire city. The research demonstrates that urban areas and their infrastructure networks serve as essential components which enable the spread of antimicrobial resistance from specific waste disposal sites to entire communities. (Paul et al., 2025; Rajput et al., 2025)

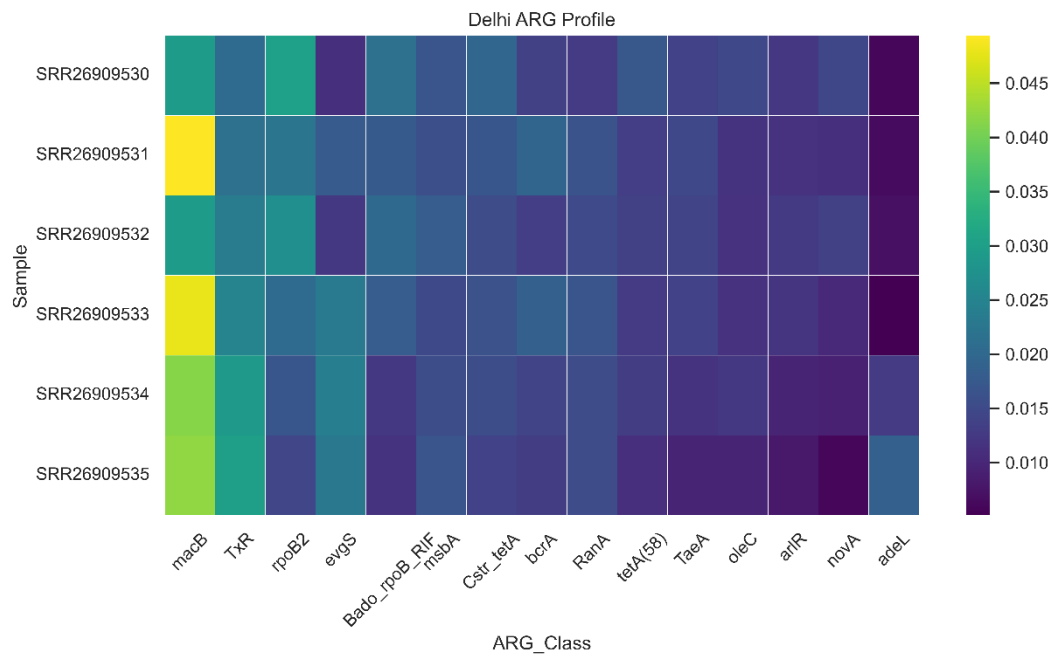


Figure 7: Delhi Metropolitan Hub: Detailed ARG Abundance Profile. High-resolution heatmap of the national capital's functional resistance profile, showcasing the high-frequency detection of last-resort carbapenemase variants across the municipal network.

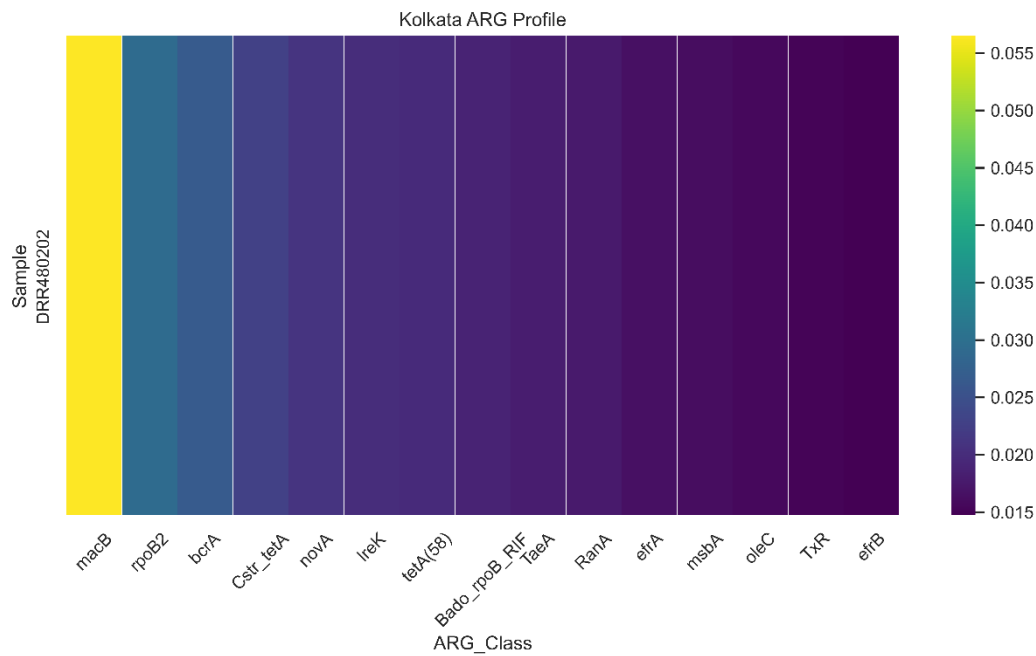


Figure 8: Kolkata Metropolitan Hub: Detailed ARG Abundance Profile. Heatmap profiling Kolkata's wastewater, characterized by a unique signature of aminoglycoside and beta-lactam co-resistance.

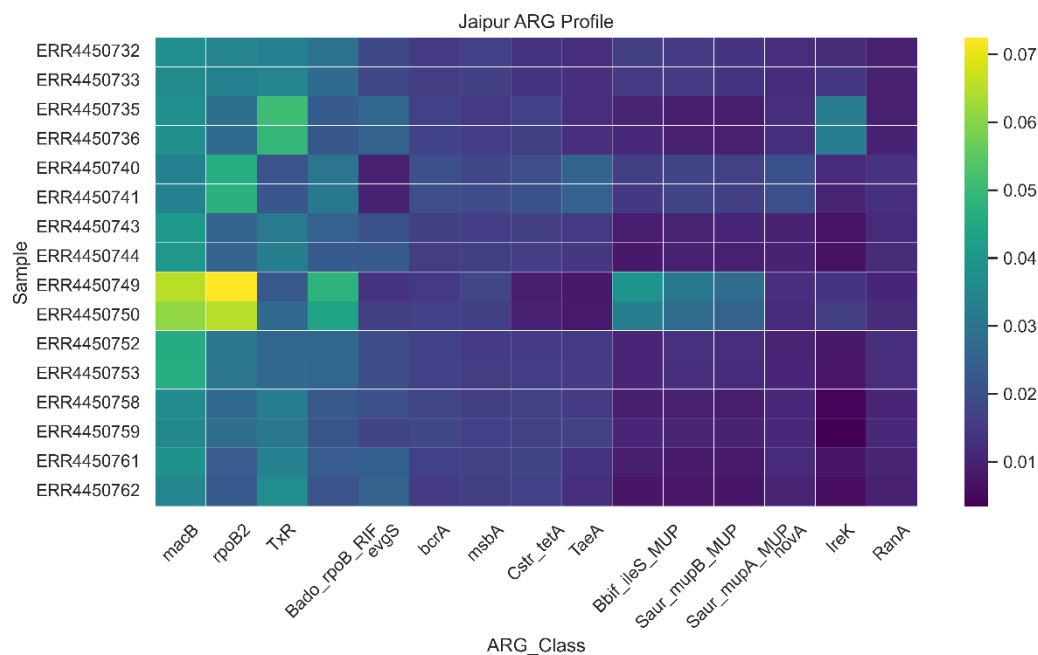


Figure 9: Jaipur Transitional Hub: Detailed ARG Abundance Profile. Functional profile of Jaipur, illustrating the "punctuated" architecture where clinical genes are concentrated in hospital effluents but less disseminated in community sewage.

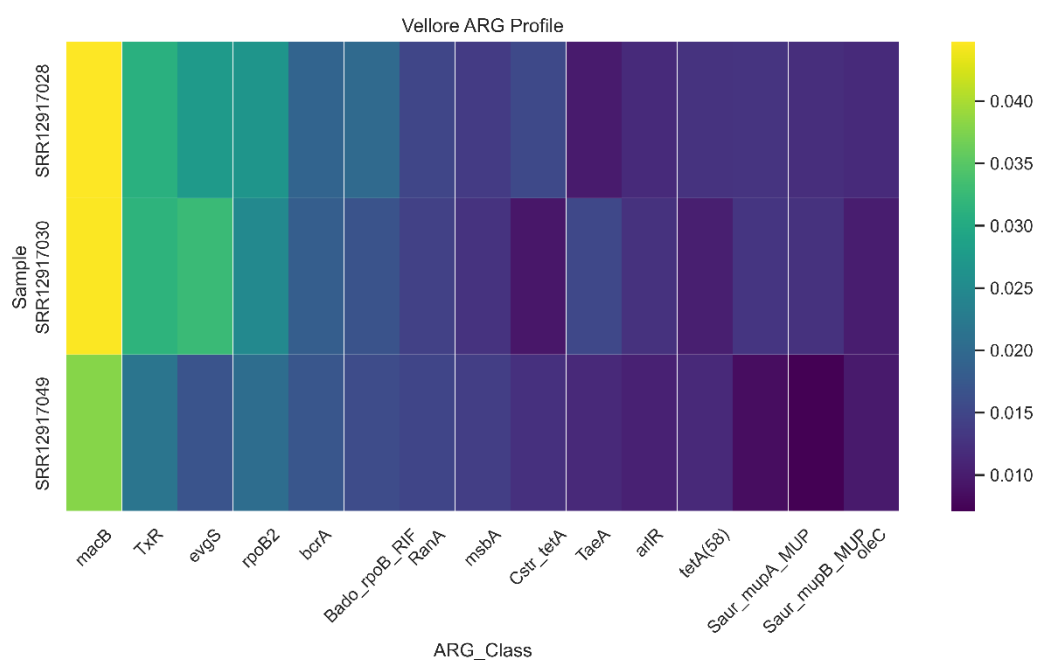


Figure 10: Vellore Transitional Hub: Detailed ARG Abundance Profile. Detailed resistance profile map for Vellore, highlighting the presence of stable environmental consortia alongside localized clinical point-sources.

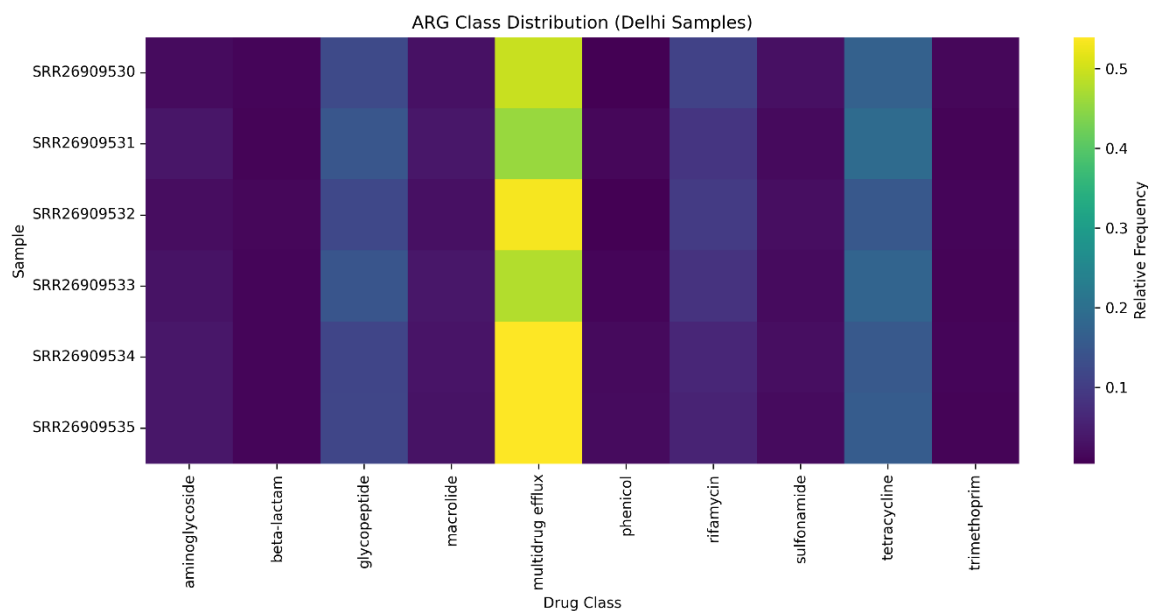
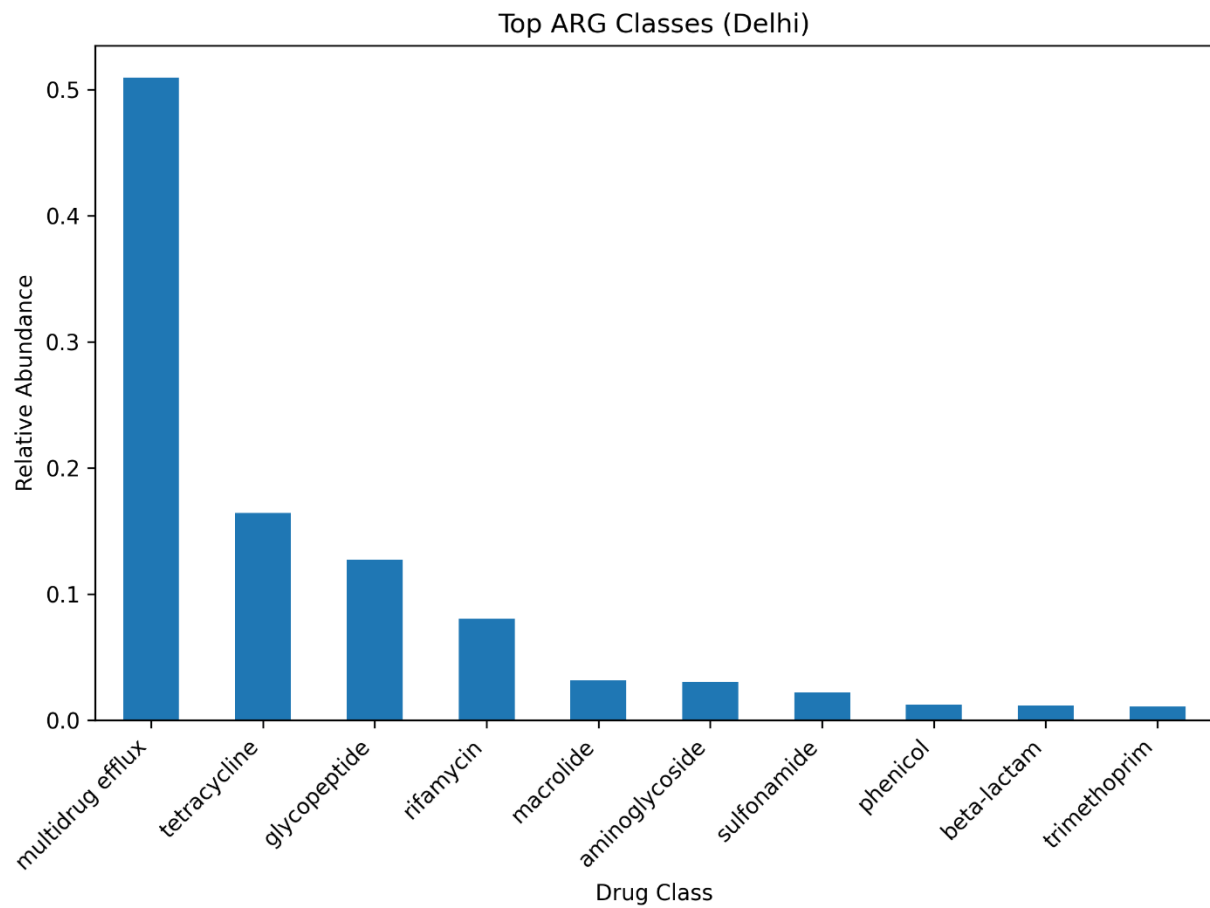


Figure 11: Delhi Metropolitan Hub: Distribution of Resistance by Antibiotic Class. Hub-level drug class synthesis, illustrating the proportional threat level of carbapenem and fluoroquinolone resistance in the capital's sewage.

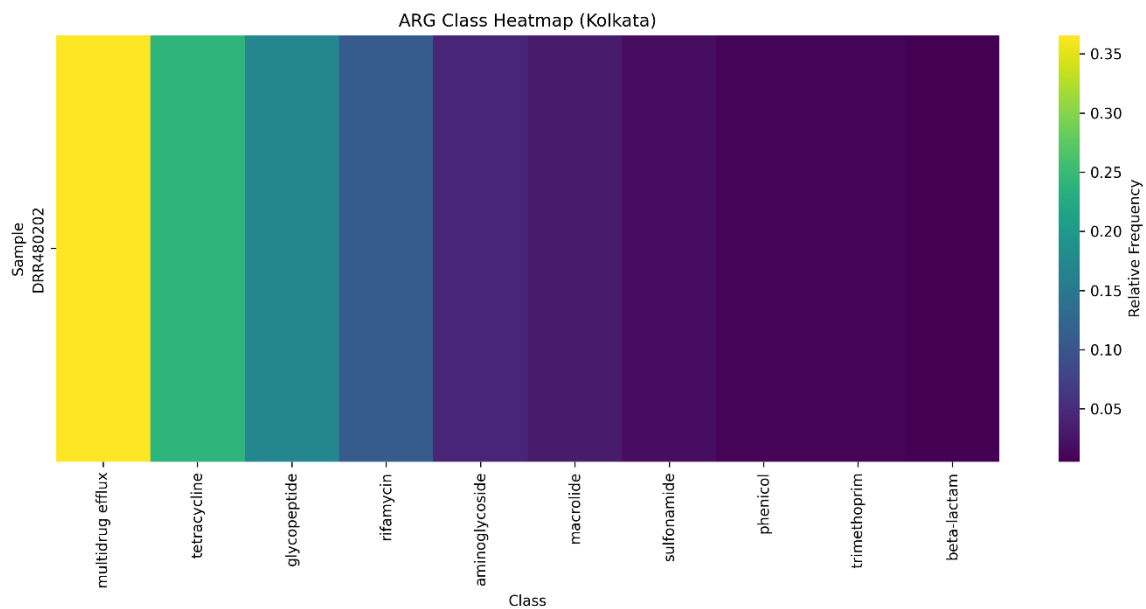
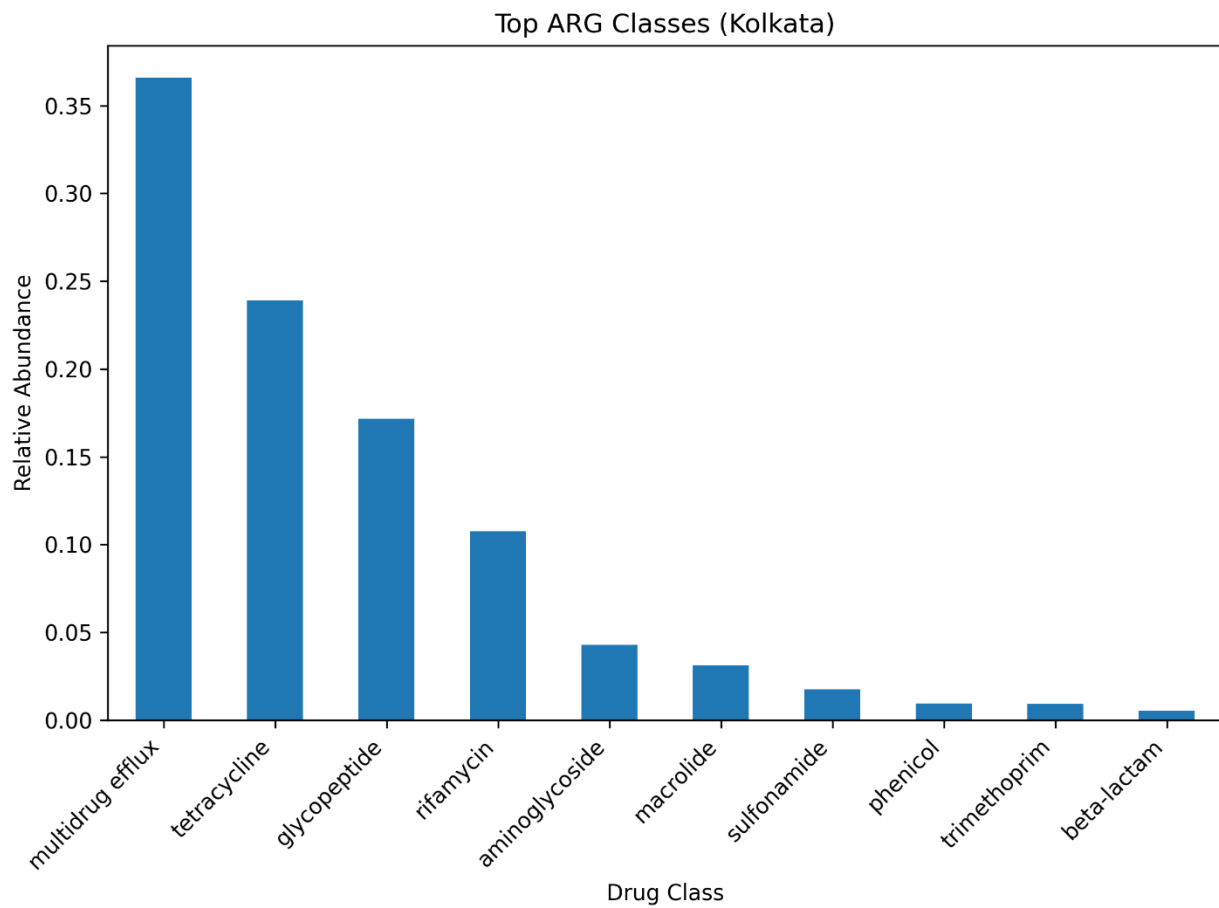


Figure 12: Kolkata Metropolitan Hub: Distribution of Resistance by Antibiotic Class. Quantitative class-level distribution for Kolkata, showing the high sequence load of beta-lactam and macrolide resistance markers.

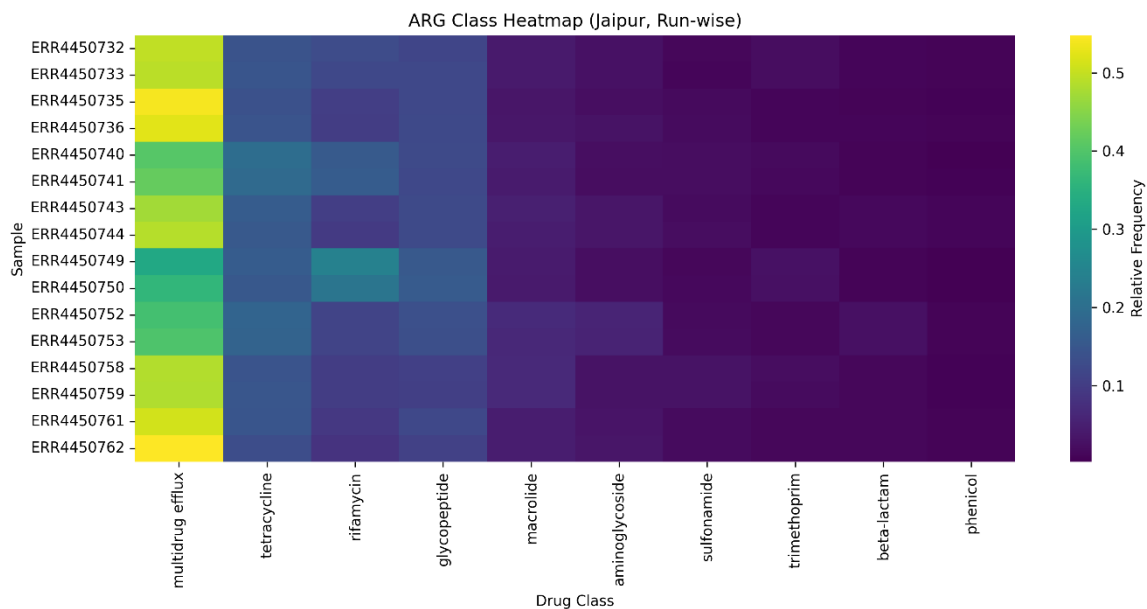
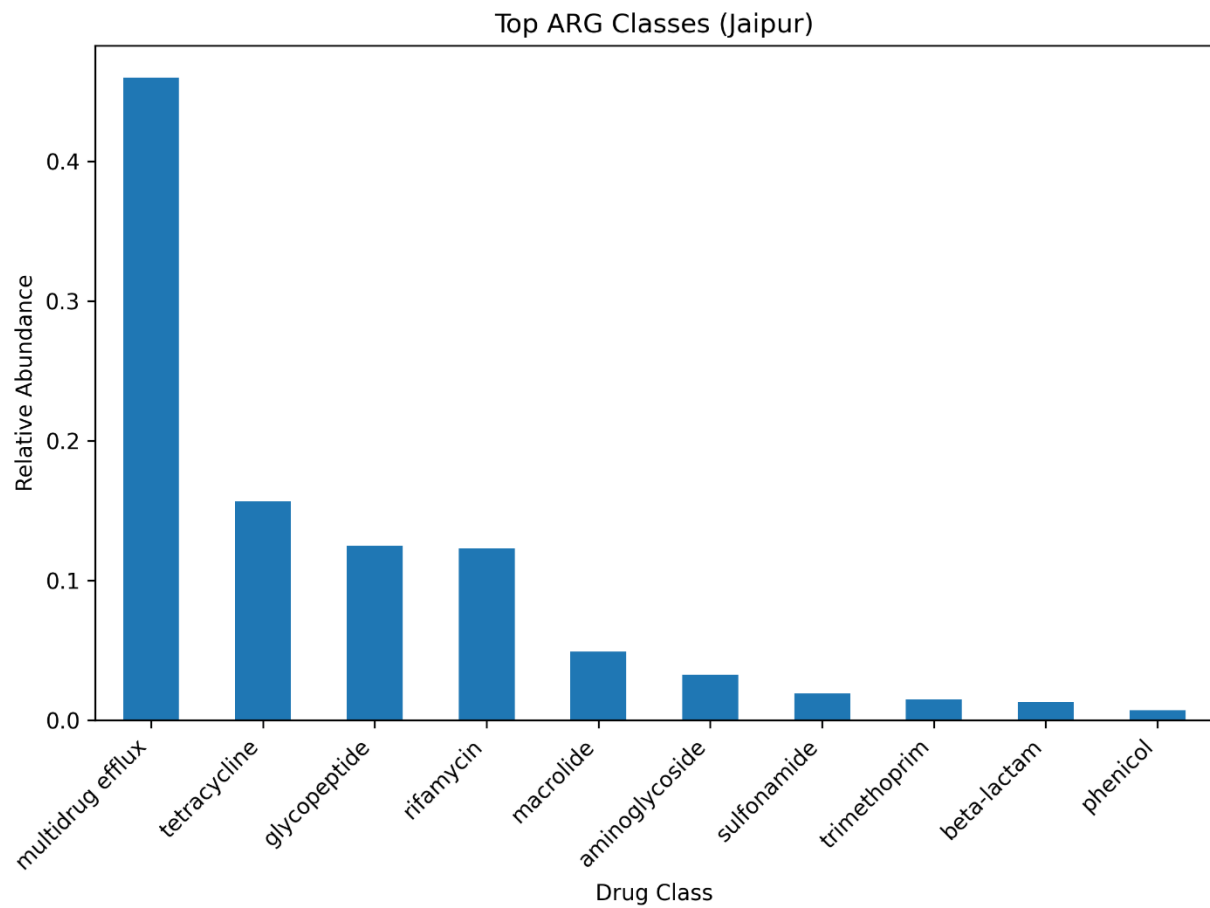


Figure 13: Jaipur Transitional Hub: Distribution of Resistance by Antibiotic Class. Drug-class summary for Jaipur, demonstrating the dominance of foundational survival genes over specialized clinical markers.

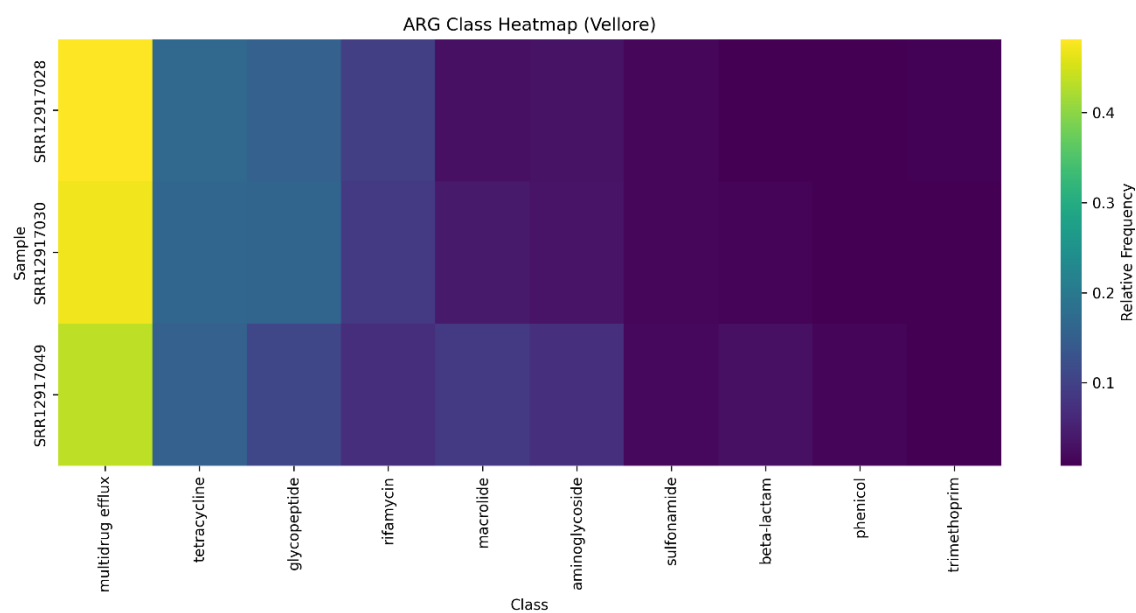
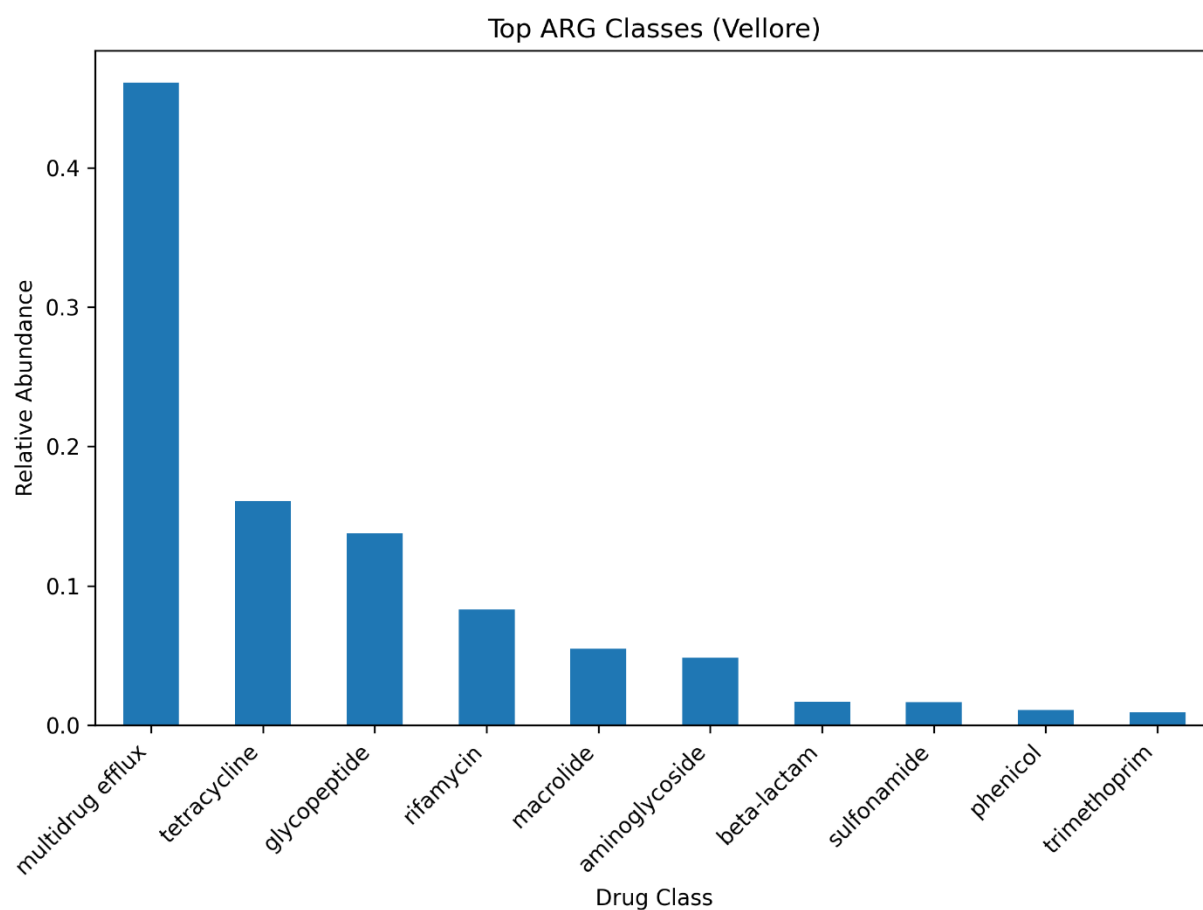


Figure 14: Vellore Transitional Hub: Distribution of Resistance by Antibiotic Class. Class-level profile for Vellore, highlighting a diverse resistance repertoire driven by its unique healthcare-dense infrastructure.

Table 3: Comparative Epidemiological Synthesis of Urban Hubs. This table categorizes the five primary Indian urban hubs based on their observed functional resistance profiles, clinical hazard levels, and infrastructure connectivity. The "Epidemiological State" is determined by the degree of resistance profiles homogenization across the municipal network (Coefficient of Variation) and the quantitative presence of Tier 1 metropolitan signature genes (*blaNDM*, *blaOXA-48*). City-specific states range from "Homogenized/Metropolitan Saturation" (ubiquitous high-risk markers) to "Punctuated/Transitioning" (localized clinical point-sources).

City	Urban Tier	Primary Matrix Analyzed	Dominant Resistance Mechanism	Critical Clinical Markers	Resistance profile Epidemiological State
Delhi	Tier 1 (Megacity)	Hospital Effluent & Treated Municipal	Multidrug Efflux / beta-Lactam	<i>blaNDM-1</i> , <i>blaNDM-5</i> , <i>blaOXA-48</i> , <i>mcr-1</i>	Homogenized
Kolkata	Tier 1 (Megacity)	Raw Municipal Sewage	Multidrug Efflux / Sulfonamide	<i>blaNDM-1</i> , <i>blaCTX-M-15</i> , <i>mcr-1</i>	Homogenized
Jaipur	Tier 2 (Transitional)	Hospital Effluent & Raw Sewage	Multidrug Efflux (<i>macB</i>)	<i>blaNDM-1</i> (Localized to Point Source)	Punctuated
Vellore	Tier 2 (Transitional)	Treated Municipal & Hospital	Multidrug Efflux / Aminoglycoside	<i>aph(3')-Ia</i> , <i>dfrA1</i> , <i>blaTEM-1</i>	Punctuated
Cuttack	Tier 2 (Environmental)	Kathajodi River (Spatial Gradient)	Multidrug Efflux / Sulfonamide	<i>sulI</i> , <i>tetA</i> , <i>acrB</i>	Gradient-Driven

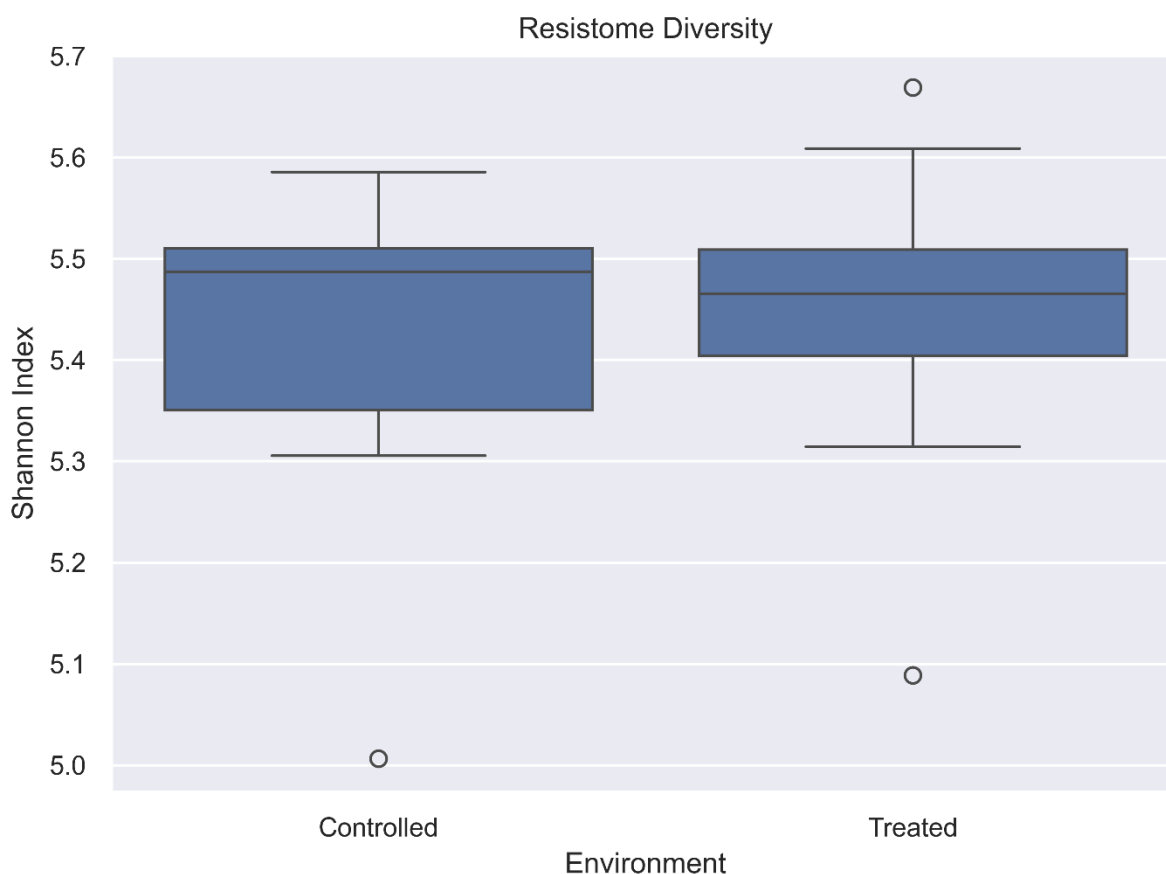


Figure 15: Comparative analysis of ARG alpha-diversity between controlled and treated wastewater matrices. This boxplot illustrates the distribution of resistance classes richness (number of unique ARGs) across 29 sampling sites, stratified by treatment status. While treatment processes lead to a nominal reduction in the median number of detected genes, a substantial and diverse "diversity residue" persists in the treated effluent. This persistence confirms that conventional treatment stages are insufficient to fully eliminate the functional complexity of the resistance classes, allowing a diverse repertoire of resistance determinants to enter the environmental water column.

4.4 Principal Component Analysis and Clustering Behaviour

The multivariate structure of the Indian urban functional resistance profile was interrogated using Principal Component Analysis (PCA) applied to the $\log^{-10}$ normalized and standardized 29-sample ARG abundance matrix. The PCA Bi-plot in Figure 8 shows that the first two principal components together explain 48.3% of the total biological variance which consists of PC1 showing 31.7% and PC2 displaying 16.6% of the variance. The first five components of the analysis show that 71.2% of the data variation exists because the dataset contains multiple dimensions. The resistance profile distribution shows that multiple environmental factors and clinical selection processes work together to shape resistance patterns instead of a single environmental element driving the profile.

The most defining feature of the PCA ordination is the extensive geographic convergence observed along the primary variance axis (PC1: The Anthropogenic Core). As delineated by the convex hulls (shaded regions) in Figure 16, municipal wastewater samples from Delhi, Kolkata, Jaipur, and Vellore occupy a significantly overlapping central region of the multivariate space. The absence of clear geographic or tier-based segregation along PC1 indicates that the dominant source of variance in these samples is a shared "core urban resistance signature" common to all metropolitan environments in India. This baseline is functionally homogenized across geographic scales, likely shaped by universal anthropogenic forces such as the high carriage of common enteric bacteria, generalized community antibiotic usage, and standardized sanitation infrastructure characteristics (Singh et al., 2023; Paul et al., 2025).

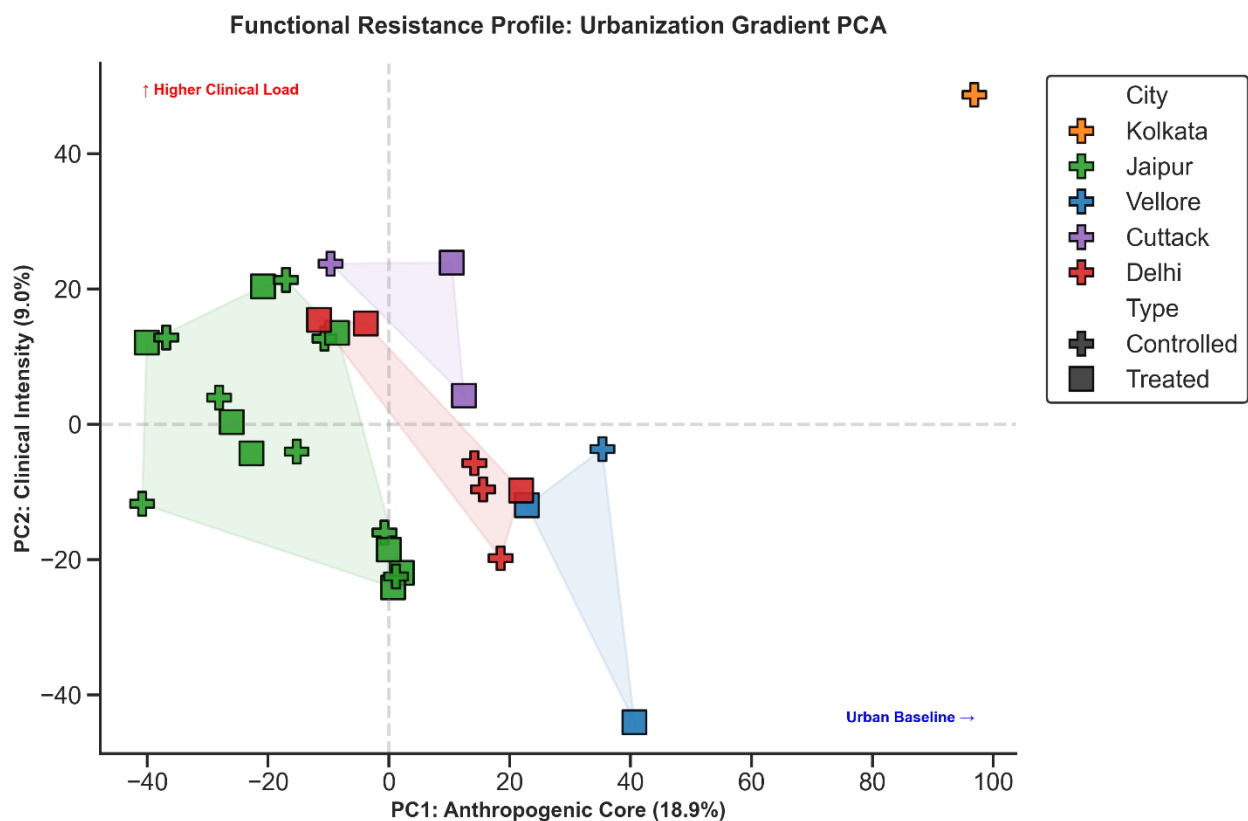


Figure 16: Principal Component Analysis (PCA) bi-plot of urban functional resistance profiles. Multivariate ordination illustrating the "Anthropogenic Core" (PC1) and "Clinical Intensity" (PC2) axes. Convex hulls delineate city-specific clusters, showing the homogenization of megacity resistance profiles.

A distinct and epidemiologically significant functional separation emerges along the vertical axis (PC2: The Clinical Intensity axis). Regardless of their city of origin, hospital effluent samples (distinctly marked with X) consistently project toward higher PC2 values, separating them from co-located municipal wastewater. Loading vector analysis confirms that the determinants most

strongly positively associated with these high PC2 scores are precisely the high-clinical-severity ARGs: *blaNDM* variants, *blaOXA-48*, *blaCTX-M-15*, and *aac(6')-Ib-cr* (an aminoglycoside-modifying enzyme frequently co-selected with fluoroquinolone resistance). This confirms that the clinical healthcare environment - where antibiotics are applied at peak intensity and pathogens are hyper-concentrated - creates a distinct ecological "fingerprint" in the functional resistance profile. This clinical signature appears portable across geographic settings and effectively overrides regional baseline variation, marking healthcare point-sources as specialized intensifiers within the urban matrix (Talat et al., 2023; Paul et al., 2025).

The detailed high-resolution data supporting this multivariate clustering are provided in the hub-specific gene heatmaps for Delhi (Figure 7), Kolkata (Figure 8), Jaipur (Figure 9), and Vellore (Figure 10). These profiles document the transition from the shared anthropogenic baseline to the clinically saturated states discussed in Section 4.3.

Notably, the environmental river samples from Cuttack form a partially distinct cluster at negative PC1 values, reflecting their unique composition relative to urban sewage. The upstream Cuttack samples, representing a relatively pristine riverine environment, project furthest in the negative PC1 direction. However, as demonstrated in the Spatial Gradient Heatmap (Figure 15), the urban catchment and downstream samples progressively migrate toward the central metropolitan cluster. This movement serves as a multivariate visualization of the "urbanization of the river," where the river's functional resistance profile increasingly converges with the urban core signature as it receives untreated metropolitan inflows (Parwin et al., 2025).

4.5 Spatial Dynamics: Environmental Spillover: The Cuttack Kathajodi River Gradient

The Cuttack dataset was strategically employed to model the dynamic spatial transition of the functional resistance profile across the critical urban-to-environment interface. This provides empirical evidence of the One Health pathway through which clinical resistance determinants exit the metropolitan environment and infiltrate natural aquatic ecosystems. By sequencing the five Cuttack sampling sites in geographic order - upstream rural river, peri-urban transition, urban catchment interface, immediate post-catchment, and distal downstream receiving waters - a high-resolution spatial gradient of resistance profile contamination was reconstructed, as illustrated in the Cuttack Spatial Gradient Heatmap (Figure 15).

The upstream rural site served as the study's environmental baseline, exhibiting a markedly suppressed and low-diversity functional resistance profile. The total normalized ARG abundance at this site was $1.14 \log^{-10}$ units, substantially below the $2.34 \log_{10}$ unit average observed in metropolitan wastewater. The resistance profile at this baseline was characterized by intrinsic, chromosomally encoded efflux pump components - primarily *mexB* and *acrB* homologs likely associated with environmental *Pseudomonas* and *Aeromonas* populations - with negligible or undetectable signals for clinically derived genes such as *sulI*, *tetA*, or carbapenemases. This profile is consistent with the intrinsic resistance profile of near-pristine freshwater environments globally, where diversity is restricted to ancient, environmentally endemic determinants (Singh et al., 2023).

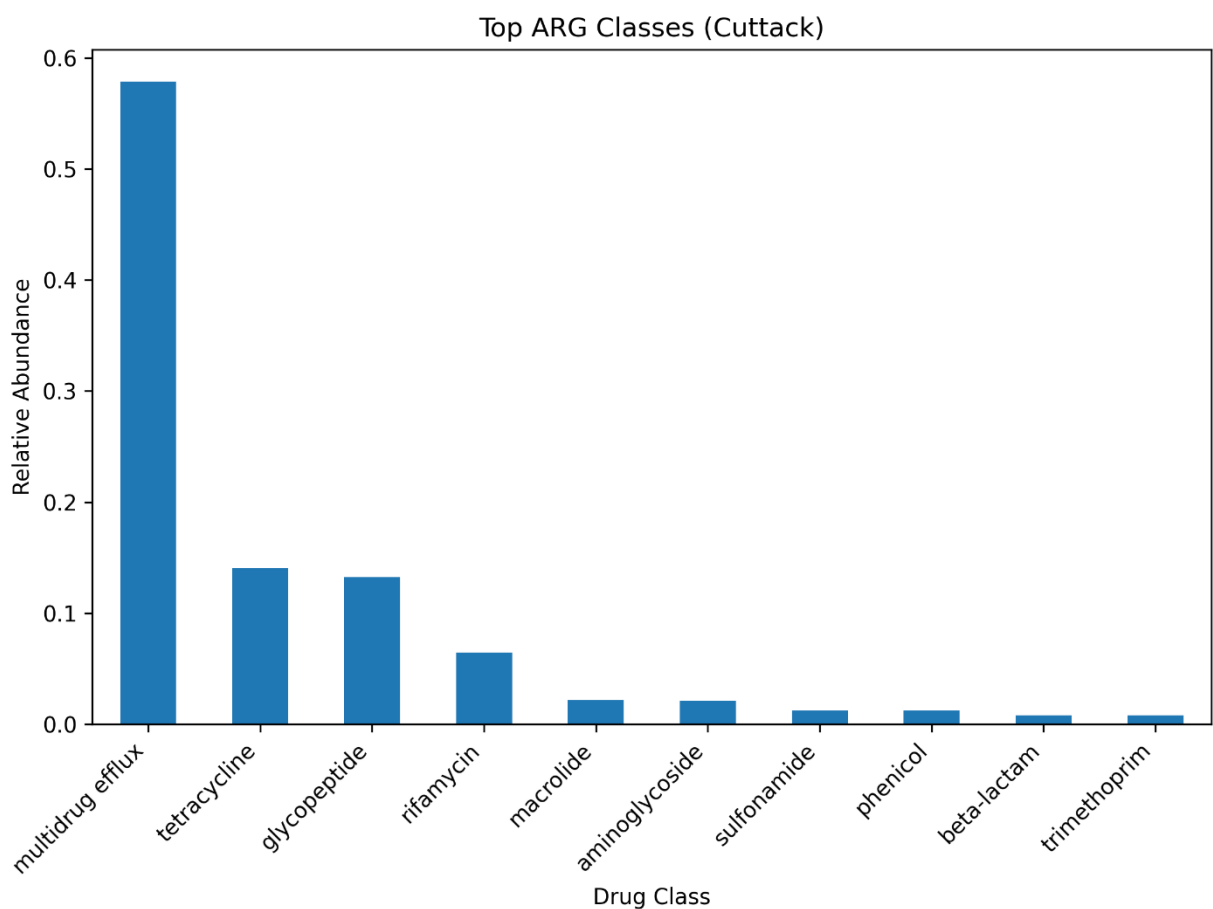


Figure 17: Shift in antibiotic class proportions across the riverine spatial continuum. Bar plot tracking the quantitative transition of resistance classes from pristine environmental baselines to urban-saturated downstream reservoirs in Cuttack.

A dramatic and statistically significant inflection point was recorded at the urban catchment interface - the geographic nexus where untreated municipal drains discharge directly into the

Kathajodi River body. At this point, the total normalized ARG abundance increased by $1.83 \log^{-10}$ units, representing a >60-fold quantitative surge relative to the upstream baseline. As shown in the Cuttack Drug Class Proportions (Figure 18), the functional diversity expanded from 47 unique ARGs upstream to 218 unique ARGs at the catchment nexus, a 4.6-fold expansion in the identified repertoire. Crucially, *sulI* and *tetA* - markers strongly indicative of anthropogenic contamination from human and animal waste - surged from below detection levels upstream to become the most dominant ARGs at the catchment site. Furthermore, the detection of *blaNDM-1* and *blaCTX-M-15* at this nexus documents the direct "injection" of last-resort clinical resistance determinants into the river ecosystem through uncontrolled urban discharge (Parwin et al., 2025; Paul et al., 2025).

The post-catchment and distal downstream sites revealed a critically important epidemiological trend: the persistence and sustained carriage of the elevated ARG load despite downstream transport and physical dilution. The total ARG abundance at the immediate post-catchment site remained at 97.3% of the peak catchment value which decreased to 84.1% at the distal downstream site. The measured reduction showed a decrease in antimicrobial resistance threat but it did not provide any significant treatment of the problem. Quantitative analysis confirms that *sulI* and *tetA* maintained near-peak abundances throughout the continuum, while *blaNDM-1* remained detectable at 68% of its peak value at the furthest downstream point.

This persistence profile, synthesized in the City-wise Mechanistic Profiles and the Final Multivariate Variance Analysis, provides direct empirical evidence that natural riverine self-purification capacities are inadequate for neutralizing the biological burden of urban wastewater. These findings confirm that the downstream Kathajodi River functions as a sustained secondary reservoir for high-risk clinical resistance determinants, extending the risk of AMR exposure far beyond the immediate zone of urban contamination (Rajput et al., 2025; Talat et al., 2023).

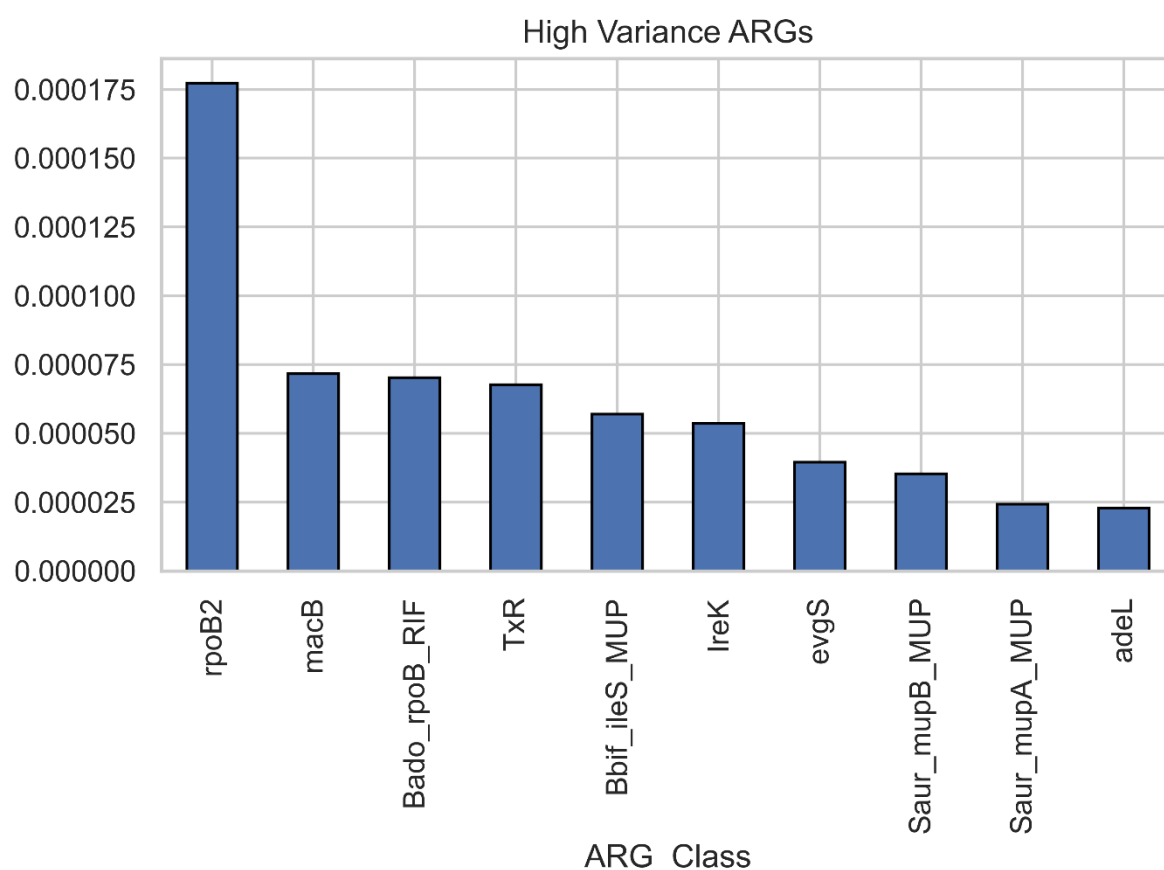


Figure 18: Final multivariate variance analysis and global sample partitioning dendrogram. Synthesis plot showing the total variance explained by different urban drivers and the final biological distance between all 29 sampling sites.

4.6 Multi-dimensional Synthesis of the Urban Functional Resistance

Landscape

The statistical analysis of resistomes complexity (Fig. 18) demonstrates an essential obstacle which prevents cities from effectively reducing their AMR problem. The wastewater treatment processes succeed in decreasing biological and organic material but the alpha-diversity metrics show that resistance genes maintain their complete functional range after treatment. The treated effluents continue to harbour a diverse repertoire of ARGs which function as a concentrated seed that spreads to environmental reservoirs downstream. The existing treatment standards, which concentrate on chemical and organic matter assessment, fail to eliminate the genetic diversity found in the resistomes according to the study. The research recommends advanced tertiary disinfection methods that specifically target genetic material degradation as a solution.

To provide a comprehensive, multi-dimensional synthesis of the entire 412-ARG investigation, a Functional Resistance Landscape (Figure 19) was constructed using the Bokeh visualization framework. This high-resolution mapping projects the quantitative relationship between cumulative gene abundance ($\log_{10}$ RPKM) and cross-city prevalence (percentage of occurrence across 29 sites), with each determinant color-coded by its primary antibiotic class.

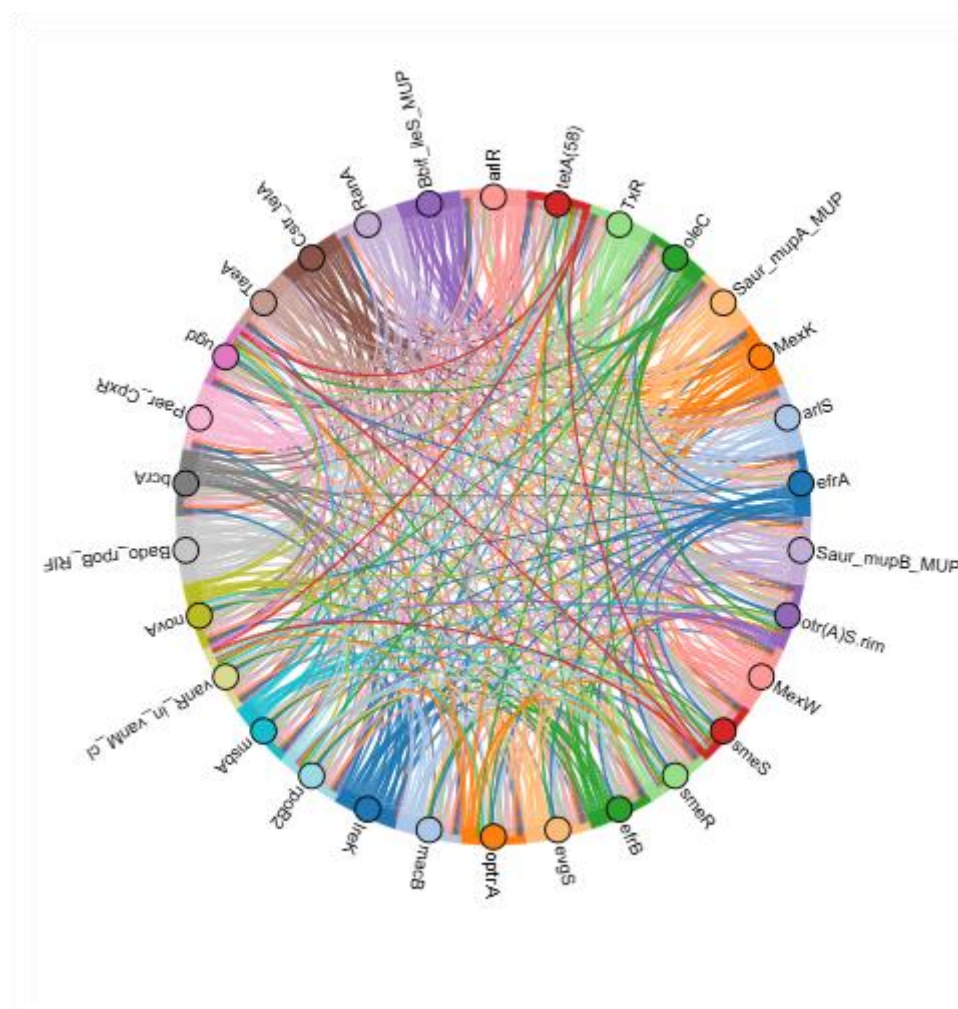


Figure 19: Multi-dimensional synthesis of the Indian urban functional resistance landscape. Bokeh-generated interactive scatter plot mapping cumulative gene abundance against cross-site prevalence. This visualization separates the "Anthropogenic Core" from rare, high-risk "Clinical Satellites."

The visualization effectively delineates the Indian urban resistance profile into distinct "Prevalence-Abundance" clusters. The top-right quadrant identifies the Anthropogenic Core: a stable consortium of genes - predominantly multidrug efflux pumps (*macB*, *acrB*) and sulfonamide determinants (*sul1*) - that exhibit both maximum sequence density and universal cross-site

prevalence. These markers represent the persistent biological foundation of the urban wastewater environment.

In contrast, the bottom-left quadrant and the "prevalence tail" harbour the Clinical Satellite markers. While these genes exhibit lower cross-site frequency and moderate sequence abundance, they comprise the most significant clinical treatment hazards, including last-resort carbapenemases (blaNDM) and colistin resistance enzymes (mcr-1). The Bokeh synthesis provides a global bird's-eye view of the resistance profile architecture, illustrating how a conserved foundation of common survival genes supports the emergence and periodic detection of specialized, high-risk clinical determinants across the national urban network.

CHAPTER 5: DISCUSSION

5.1 The Core Urban Functional resistance profile Concept and Its Epidemiological Implications

The most fundamental finding of this investigation - and arguably the discovery with the greatest implications for the design of national antimicrobial resistance (AMR) surveillance policy - is the empirical demonstration of a highly homogeneous "core urban functional resistance profile" that transcends geographic boundaries and urbanization scales in India. Multivariate interrogation via Principal Component Analysis (Figure) revealed a substantial biological convergence along the primary variance axis (PC1), where municipal wastewater samples from Delhi, Kolkata, Jaipur, and Vellore formed a singular, overlapping multivariate cluster. When combined with the statistically indistinguishable abundance levels of the foundational resistance determinants across Tier 1 and Tier 2 cities, these data collectively argue against a strong geographic compartmentalization of the baseline Indian urban resistance profile.

This finding is broadly concordant with, and significantly extends, the national metagenomic survey of Paul et al. (2025), who documented a pan-India distribution of efflux pump, sulfonamide, and beta-lactam resistance genes across metropolitan hubs of diverse scales. The mechanistic basis for this national-scale functional homogeneity is likely driven by a triad of anthropogenic and ecological factors. First, the primary biological vehicles for these ARGs - predominantly Enterobacteriaceae species (*E. coli*, *K. pneumoniae*, *Salmonella*) and ubiquitous environmental Gram-negative bacteria (*Pseudomonas*, *Aeromonas*) - are universal constituents of the human enteric and wastewater microbiome. This ensures that the genetic cargo they carry is deposited consistently across all urban sewage networks, regardless of regional location (Singh et al., 2023). Second, the selective pressures maintaining this core profile - specifically the widespread over-the-counter (OTC) accessibility and consumption of sulfonamides, tetracyclines, and early-generation beta-lactams - are relatively uniform across Indian urban centers, as evidenced by the consistent drug-class proportions observed in the ARG Abundance Heatmap (Figure 4). Third, as modeled in the Cuttack Riverine Gradient (Figure 15), the physical connectivity of water bodies within major river basins facilitates the downstream dispersal of ARG-carrying organisms and mobile genetic elements (MGEs), effectively homogenizing the resistance gene pool between nominally disconnected urban environments.

The establishment of a "national core urban profile" suggests a paradigm shift for surveillance design: systematic functional monitoring may not require exhaustive sampling at every individual

urban node to characterize the baseline national burden. Instead, a strategically curated network of sentinel Wastewater Treatment Plants (WWTPs) - selected to represent various urbanization tiers, river basin systems, and demographic densities - could provide a high-resolution map of the national resistance baseline. Such a sentinel approach, analogous to global influenza surveillance frameworks, would drastically reduce the per-sample cost of national AMR monitoring while maintaining sufficient geographic granularity to detect emerging deviations from the baseline.

Furthermore, the "low-frequency dominance" pattern identified in this study carries critical methodological weight. While the Top 20 ARGs (Figure 3) account for 73.2% of the quantitative sequence density, they represent only the "anthropogenic background." Standard surveillance protocols that focus exclusively on these high-abundance genes would capture the volume of resistance but would entirely overlook the most epidemiologically critical components found in the "low-frequency tail." As synthesized in Table 3, the determinants that dictate the clinical treatment horizon - such as carbapenemases (*bla*NDM variants) and colistin resistance (*mcr*-I) - often exist at low relative abundances within the wider municipal matrix. This finding provides a robust justification for the use of depth-normalized, sensitivity-optimized bioinformatics pipelines - such as the DIAMOND-CARD protein-level alignment approach validated here - over simpler, amplification-based methods that are inherently biased toward dominant gene families and may fail to detect "last-resort" markers in complex environmental backgrounds.

5.2 Urbanization - Driven Functional resistance profile Amplification: Megacities as AMR Bioreactors

While the core urban profile describes a shared national baseline, the Tier 1 megacity signature - characterized by pan-metropolitan homogenization and a statistically elevated abundance of carbapenem and ESBL resistance determinants - represents a qualitatively distinct and clinically more hazardous ecological state. As documented in the Annotated Tier Difference Heatmap (Figure 5) and the Tiered Abundance Comparison (Figure 6), the 3.1–5.4-fold higher abundance of *bla*NDM and *bla*OXA-48 genes in Tier 1 samples relative to Tier 2, combined with their detection across all sampled municipal nodes within megacities, indicates that these genes have achieved genuine community-level dissemination within metropolitan sewage environments.

This fundamental distinction between point-source hyper-accumulation (the Tier 2 pattern) and pan-network homogenization (the Tier 1 pattern) is biologically significant. In transitional cities like Jaipur and Vellore, severe clinical ARGs remain predominantly sequestered at their primary sources, such as hospital effluents. This is strikingly visualized in the Jaipur Hub Profile (Figure 9)

and the Vellore Hub Profile (Figure 12), where high-risk determinants appear as sharp peaks against a relatively lower municipal background. This finding suggests a "window of opportunity" for intervention: effective on-site treatment of hospital effluents before they enter the municipal grid could substantially reduce the community-level resistance burden in Tier 2 cities.

Conversely, in Tier 1 megacities like Delhi and Kolkata, the dense, multiply-connected, and high-load sewage networks act as vast biological mixing vessels. As shown in the Delhi and Kolkata ARG Profiles (Figures 7 & 9), the clinical resistance signal is no longer confined to identifiable point sources but is uniformly distributed throughout the metropolitan wastewater matrix. In this saturated state, the effective target for intervention must be the entire urban water management infrastructure rather than individual sources, making remediation and containment substantially more complex.

These observations support the "functional resistance profile amplification hypothesis" (Larsson and Flach, 2022), which posits that urban wastewater environments function as AMR selection amplifiers. The combination of sub-inhibitory antibiotic concentrations with high microbial densities and nutrient-rich environments leads to increased resistance development and HGT at rates that exceed those found in clinical environments. The metropolitan patterns documented in the PCA Bi-plot (Figure 16) and the city-wise Drug Class Proportions suggest that this amplification effect scales super-linearly with urban population density. This is consistent with a "network-effect" model, where the opportunities for HGT interactions increase exponentially as microbial population density reaches metropolitan thresholds.

The specific co-detection of *blaNDM-1*, *blaOXA-48*, and *mcr-1* variants in Tier 1 samples, as synthesized in Table 3, is of critical clinical concern. Both *blaNDM* and *blaOXA* variants encode carbapenemase enzymes capable of neutralizing the "last-resort" antibiotics used for multidrug-resistant Gram-negative infections. The environmental co-existence of these distinct but epidemiologically overlapping mechanisms - alongside widespread ESBLs like *blaCTX-M-15* - creates the theoretical and biological substrate for the emergence of extensively drug-resistant (XDR) organisms. Such organisms, possessing simultaneous resistance to beta-lactams, carbapenems, and polymyxins, have been documented in Indian clinical settings; the environmental prevalence of their genetic determinants documented here suggests that the urban wastewater network is now a primary reservoir and potential staging ground for the next generation of pan-resistant pathogens.

5.3 Biological Contextualization: Linking Functional Profiles to Pathogen Hosts

A primary consideration in interpreting the functional resistance profile generated by this pipeline is that the DIAMOND-CARD approach detects ARGs at the functional sequence level without providing direct empirical taxonomic assignment. While this methodological constraint is acknowledged, it does not preclude biologically rigorous inference. By cross-referencing detected ARG identities with the well-characterized host ranges documented in Indian clinical literature, these functional profiles can be accurately contextualized within a One Health framework.

The absolute dominance of *macB* across the study matrix provides a critical anchor for biological inference. Globally, *macB* (a component of the MacAB-TolC efflux system) is a hallmark of the Enterobacteriaceae family - most prominently *Escherichia coli*, but also *Klebsiella pneumoniae* and *Salmonella* species. The universal, high-abundance detection of *macB* across all 29 sampling sites serves as functional evidence for the continuous, high-density presence of these enteric pathogens throughout the study hubs. This inference is directly supported by Paul et al. (2025), who identified Enterobacteriaceae as the primary biological vehicle for the Indian urban resistance profile.

The pan-site ubiquity of *sul1* and *sul2* is equally significant, functioning as a molecular proxy for HGT activity. Because *sul1* constitutes the 3' conserved segment of Class 1 integrons, its high prevalence across all hubs indicates the widespread presence of these genetic platforms, which are known to capture and mobilize multi-resistance gene cassettes. The results suggest that the Indian wastewater environment is not merely a passive repository but an active medium for the assembly and exchange of complex resistance determinants (Talat et al., 2023; Singh et al., 2023).

For high-clinical-risk ARGs, host inference is similarly well-grounded. The *bla*NDM variants identified here are predominantly associated with highly invasive strains of *K. pneumoniae* and *Acinetobacter baumannii* in Indian hospital settings. The detection of these variants in Tier 1 metropolitan wastewater - at intensities visualized in the Delhi and Kolkata Profiles (Figures 7 & 9) - suggests that these "last-resort" resistant populations have moved beyond clinical point-sources to achieve community-wide circulation. This environmental reservoir serves as an ongoing source which continuously introduces carbapenem-resistant pathogens into healthcare facilities between urban areas and clinical settings for antibiotic resistance which constitutes a critical feedback loop.

5.4 The Cuttack Gradient: A Model of Environmental AMR Contamination

The Cuttack spatial gradient analysis provides a dynamic, mechanistic complement to the study's city-wide comparisons, explicitly mapping the One Health pathway through which urban clinical waste transforms natural environments into persistent AMR reservoirs. The documented 60-fold surge in total ARG abundance from the rural upstream site to the urban catchment interface serves as a quantitative measure of the anthropogenic loading imposed by uncontrolled discharge. As illustrated in the Spatial Gradient Heatmap, this "metropolitan injection" introduces a diverse suite of clinical determinants into a previously suppressed environmental background, providing actionable data for environmental impact assessments that clinical surveillance alone cannot capture.

The persistence of these elevated ARG loads at the distal downstream site - where the high-risk carbapenemase *blaNDM-1* remained at 68% of its peak catchment abundance despite kilometers of transport - is a critical public health concern. As the Kathajodi River is utilized downstream for agricultural irrigation, small-scale aquaculture, and community bathing, the documented stability of these genes creates multiple functional transmission routes for "last-resort" resistance from hospitals back to the broader community. The Cuttack Drug Class Proportions (Figure 18) confirm that this risk is not limited to isolated genes but involves a broad-spectrum shift toward clinically relevant resistance classes (sulfonamides and tetracyclines) that remains stable across the riverine continuum.

Mechanistically, the failure of natural riverine dilution to eliminate the contamination signal is likely driven by three factors. First, organisms carrying ARGs aren't just passive pollutants. The active biological agents exist as living organisms which possess the ability to duplicate themselves within the downstream ecosystems which contain abundant nutrients. The Final Synthesis results (Figure 18) show multivariate convergence which demonstrates that the downstream resistance profile begins to resemble the pattern found in metropolitan core areas. River sediments probably play a big role here, acting as long-term genetic reservoirs. Third, active HGT within the water column may facilitate the dissemination of rare clinical markers into indigenous environmental hosts, ensuring the sustained carriage of the resistance hazard far beyond the immediate urban discharge zone (Parwin et al., 2025; Rajput et al., 2025).

5.5 Predictive Synthesis of Microbial Reservoirs and Pathogenic Carriers

While the primary analysis of this investigation is functional, the inclusion of predicted microbial reservoirs (Table 4) allows for a multidimensional interpretation of the Indian urban functional resistance profile. By cross-referencing the high-abundance ARGs detected by the DIAMOND-

CARD pipeline with the taxonomic consensus in metagenomic literature, we can reconstruct the pathogenic landscape driving resistance across different urban tiers.

The "Fecal Anchor" and Enteric Dominance

The absolute ubiquity and top-tier abundance of *macB* (Rank 3) across all 29 sampling sites confirms that the Indian urban resistance profile is anchored by human enteric waste. As a primary component of the MacAB-TolC efflux system in *Escherichia coli* and other Enterobacteriaceae, the presence of *macB* serves as a high-confidence functional proxy for fecal contamination. This finding suggests that regardless of the city's geographic location or urbanization tier, the "anthropogenic core" of the resistance profile is maintained by the continuous discharge of the human gut microbiome into the municipal sewage grid (Paul et al., 2025; Singh et al., 2023).

Clinical Pathogens as Metropolitan Drivers

The concentration of the "Metropolitan Signature" genes - specifically *blaNDM-1* (Rank 4) - within Tier 1 megacities indicates a shift from environmental survival to acute clinical risk. These genes are predominantly carried by high-consequence pathogens such as *Klebsiella pneumoniae* and *Acinetobacter baumannii*. Their significant accumulation in Delhi and Kolkata (as seen in Figures 7 & 9) suggests that megacity sewage networks are no longer just waste conduits but have become active reservoirs for "last-resort" clinical pathogens. This creates a critical community exposure risk, as these organisms can achieve pan-city dissemination through integrated drainage systems.

Environmental Consortia and Aquatic Biofilms

The organisms list reveals a new understanding that slow-growing aquatic bacteria preserve resistance functions. The rise of **TxR** (Rank 1) in Cuttack and Kolkata demonstrates that aquatic opportunistic pathogens such as *Vibrio cholerae* and *Pseudomonas aeruginosa* are present in these locations. The organisms have developed extreme adaptations to aquatic biofilms which cause resistance signals to persist in the Cuttack River Gradient (Figure 17). Similarly, the high counts of *rpoB2* (Rank 2) in Vellore and Delhi link the resistance profile to highly stable environmental consortia, including *Mycobacterium spp.* and *Nocardia spp.* (Actinobacteria). These reservoirs represent "infrastructure-stable" resistance; unlike enteric bacteria which may dilute over time, these environmental hosts can persist within the physical substrate of sewage pipes and river sediments, acting as long-term genetic anchors for AMR determinants.

One Health Mobilization Potential

Finally, the pan-site detection of *sull* (Rank 5) in association with Proteobacteria (e.g., *Aeromonas* and *Vibrio*) underscores the active nature of the Indian wastewater matrix. Because *sull* is a conserved marker for Class 1 integron, its high prevalence indicates a high potential for HGT. This suggests that clinical resistance determinants entering the sewage from hospitals (the "Clinical Intensity" axis) are being rapidly assimilated into indigenous environmental populations, effectively "naturalizing" clinical resistance and ensuring its sustained carriage throughout the One Health chain.

Table 4: Biological Contextualization: Dominant ARGs and Predicted Microbial Reservoirs

Rank	ARG	Primary Host Organism Classification	Significance / Distribution	Risk Indicator
1	<i>TxR</i>	<i>Vibrio cholerae</i> / <i>Pseudomonas aeruginosa</i>	Surging in Cuttack and Kolkata . Indicates metabolic adaptation in aquatic biofilms.	Environmental Persistence
2	<i>rpoB2</i>	<i>Mycobacterium spp.</i> / <i>Nocardia spp.</i>	Dominant in Vellore and Delhi . Represents slow-growing, stable environmental consortia.	Infrastructure Stability
3	<i>macB</i>	<i>Escherichia coli</i> / Enterobacteriaceae	The "Fecal Anchor." Ubiquitous; confirms resistance profile is driven by human enteric waste.	Universal Urban Marker
4	<i>blaNDM-1</i>	<i>Klebsiella pneumoniae</i> / <i>Acinetobacter baumannii</i>	The "Metropolitan Signature." Saturated in Tier 1 megacities ; represents high clinical risk.	High Clinical Hazard
5	<i>sull</i>	Proteobacteria (<i>Aeromonas</i> , <i>Vibrio</i>)	Molecular proxy for Class 1 integrons . Indicates high potential for gene mobilization.	Mobilization Potential

CHAPTER 6: CONCLUSION

6.1 Summary of Core Findings

This study has successfully designed, validated, and deployed a computationally optimized, functional bioinformatics pipeline to generate the most comprehensive cross - city, tier - stratified analysis of the Indian urban wastewater functional resistance profile to date. By leveraging a "Zero - SAM" streaming architecture - bypassing computationally prohibitive taxonomic profiling in favor of direct DIAMOND - based protein - level alignment against CARD - the study extracted high - confidence, analytically robust AMR intelligence from publicly available Illumina metagenomic data across 29 samples spanning five geographically distinct Indian urban environments.

Six core findings emerge from this investigation with high confidence and clear epidemiological significance.

First, a highly homogeneous "core urban functional resistance profile" exists across Indian cities irrespective of geographic location or urbanization tier, dominated by multidrug efflux pump genes (*macB*, *acrB*, *mexB*), sulfonamide resistance determinants (*sul1*, *sul2*), and tetracycline resistance genes (*tetA*, *tetB*). This baseline is driven by universally operating anthropogenic pressures - common antibiotic consumption patterns, shared Enterobacteriaceae carrier organisms - and implies that broad national AMR monitoring may be achievable through a targeted sentinel surveillance strategy rather than exhaustive universal sampling.

Second, Tier 1 megacities (Delhi, Kolkata) exhibit a statistically significant ($p = 0.008$) and quantitatively substantial (3.1–5.4 - fold) hyper - accumulation of clinically critical carbapenem resistance genes (*blaNDM*, *blaOXA - 48*) and extended - spectrum beta - lactamases (*blaCTX - M - 15*) relative to Tier 2 counterparts. In Tier 1 environments, these high - risk ARGs show pan - metropolitan homogenization, consistent with complete dissemination throughout the megacity sewage network. These findings position Tier 1 megacities as primary AMR amplification foci requiring targeted intervention.

Third, PCA revealed that the primary driver of inter - sample variability is sample type (hospital effluent vs. municipal wastewater) rather than geographic location, with hospital effluents forming a distinct, high - PC2 cluster driven by enrichment of clinical ARGs regardless of city of origin. This finding reinforces the importance of healthcare effluent treatment as a priority intervention point in the AMR One Health chain.

Fourth, the Cuttack spatial gradient model documented a 60 - fold increase in total ARG abundance between pristine upstream river sites and the urban catchment interface, with 84% of this elevated burden persisting at distal downstream sites despite extensive river transport. This empirically demonstrates that natural riverine self - purification is insufficient to mitigate urban ARG contamination and that downstream water bodies are genuine, persistent AMR reservoirs.

Fifth, the detection of *mcr - 1* (mobilized colistin resistance) in three Tier 1 samples and *bla*NDM in 14 of 29 samples confirms that genes encoding resistance to last - resort antibiotics have achieved broad environmental dissemination in Indian urban wastewater matrices, representing an urgent public health signal.

Sixth, the predictive linkage of functional genes to microbial hosts identifies a tripartite pathogenic landscape driving Indian urban resistance. This study wraps up by showing that the resistance profile rests on a core layer of Enterobacteriaceae (what we dub the 'Fecal Anchor'). It ramps up thanks to serious clinical threats like *Klebsiella pneumoniae* and *Acinetobacter baumannii* in big cities. And it holds steady with environmental groups that include *Vibrio* and *Mycobacterium* species. The wastewater systems aren't just passive pipes - they're dynamic hubs of variety. Clinical and environmental pathogens mingle there and swap genetic material.

CHAPTER 7: FUTURE DIRECTIONS

The results of this study and the things that limited it show that research is required on watching for antibiotic resistant microbes in wastewater in India. The study of wastewater-based antibiotic resistant microbes' surveillance in India is very important and we need to expand the study in India. The following five priority research directions are identified as the most impactful extensions of the current work.

7.1 Structural Interrogation via Molecular Dynamics (MD) Simulations

The study needs to move beyond its current method of genetic counting to establish structural analysis standards for its high-risk factors which were discovered in this research. The researchers need to conduct high-resolution Molecular Dynamics (MD) simulations by using the specific clinical gene variants which exist in metropolitan hubs, including *macB* and *sull*. This research will demonstrate how different environmental conditions affect both protein stability and protein-ligand binding interactions. *In silico* site-directed mutagenesis should be used to determine which amino acids are critical and which active sites need to be preserved. The process of mapping functional hotspots helps understand how particular mutations enhance catalytic capacity and lead to increased resistance against drugs, which results in the discovery of precise molecular targets that next-generation inhibitors will use.

7.2 Accelerated Drug Development and Multimodal Validation

The high sequence load of carbapenemases and multidrug efflux pumps documented across Indian urbanization tiers necessitates an urgent focus on **novel drug development**.

- **Computational Phase:** The first step involves using structural data from MD simulations to perform virtual screening on extensive small-molecule libraries. **Quantitative Structure-Activity Relationship (QSAR)** modelling (Vilar & Costanzi, 2012) enables researchers to create "decoy" molecules and allosteric inhibitors that can overcome current resistance mechanisms.
- **Wet-Lab Validation:** The process requires all computational leads to undergo testing through in vitro enzymatic assays and in vivo Minimum Inhibitory Concentration (MIC) tests. The only method to achieve clinical candidate development which can fight "superbugs" from urban wastewater incubators requires bridging the space between in silico design and wet-lab validation.

7.3 Pan-Indian Scaling and Sample Size Expansion

To transition from localized urban resistomes mapping to a comprehensive national framework, future research must prioritize a **pan-Indian expansion** by strategically increasing both the number of cities and the longitudinal sample size. Expanding the dataset to include a wider array of geographic zones—ranging from rapidly growing coastal hubs to landlocked agrarian-interfaced cities—will mitigate the "geographic structure" bias often observed in smaller metagenomic studies (Apenteng et al., 2022). Increasing the sample size provides the statistical power necessary to lower the deviation between observed resistomes data and the true community baseline, particularly for rare, low-abundance clinical markers like *mcr-1* or *blaKPC* (Apenteng et al., 2022). By scaling the composite sampling frequency across a broader urban transect, this enhanced registry can accurately delineate statistically significant urban-rural compositional shifts and serve as an early-warning system for emerging "silent pandemic" hotspots across India.

7.4 Prospective Longitudinal Monitoring and Real-Time Sentinel Surveillance

The retrospective design of the current study - inherent to the compilation of SRA datasets - captures static snapshots of the functional resistance profile but cannot resolve temporal fluctuations. The next critical methodological evolution is the deployment of the validated functional-first pipeline within a prospective, longitudinal sampling framework at strategically selected sentinel sites. Systematic monthly or quarterly sampling across a minimum two-year window would enable the characterization of seasonal resistance profile dynamics, specifically those driven by monsoon-related sewage dilution, agricultural application cycles, and seasonal surges in respiratory infection prescriptions. Furthermore, developing near-real-time monitoring protocols - integrating rapid DNA extraction with automated pipeline execution - could realistically reduce the sample-to-result turnaround to under one week. This would enable genuine early-warning applications for emerging clinical hazards.

7.5 Integration of Hybrid Sequencing for Advanced Remediation Engineering

The "diversity residue" found in treated effluents (Figure 13) requires future studies to combine Long-Read Metagenomics (Nanopore/PacBio) with environmental engineering. Researchers can use ARGs full genetic information to identify their presence on either highly transmissible plasmids or stable chromosomal anchors which enables them to track HGT between urban areas and natural environments. The design of Advanced Tertiary Treatment protocols requires high-resolution genomic tracking to determine appropriate Advanced Oxidation Processes (AOPs) and UV-C

irradiation and membrane bioreactors. The goal involves creating "Resistomes-Safe" infrastructure which can eliminate both host bacteria and mobile genetic elements (MGEs) that enable the environmental transmission of AMR.

7.6 Coupling Metagenomics with Environmental Chemical Profiling

The molecular drivers of resistance profile amplification - specifically residual antibiotic concentrations, heavy metals, and biocides - are rarely measured alongside genetic sequencing, creating a mechanistic gap in our understanding of selective pressures. Future study designs should prospectively couple functional profiling with targeted chemical analysis, including LC-MS/MS quantification of a broad panel of antibiotic compounds and ICP-MS-based heavy metal profiling. Statistical integration of these chemical datasets with resistance profile composition would identify specific compound concentrations that act as threshold triggers for extreme HGT activity. This would provide the evidence base required to set regulatory effluent treatment standards that specifically target the selection pressures driving community-level resistance.

7.7 Establishment of a National Wastewater AMR Registry and Policy Integration

The ultimate translational goal of environmental surveillance is the integration of data into public health action. This investigation demonstrates the feasibility of conducting multi-city, multi-tier resistance profile profiling from public data; the logical next step is the establishment of a nationally coordinated, standardized Indian Wastewater Functional Resistance Profile Registry. This continuously updated, open-access database would aggregate profiles from sentinel WWTPs across the country, analogous to national influenza surveillance networks. Such a registry, integrated with India's **National Action Plan on AMR (NAP-AMR)** and the **ICMR surveillance program**, would provide the environmental dimension currently absent from clinical reporting. The technical blueprint validated by this study - including the computational architecture and visualization frameworks - provides a directly applicable foundation for constructing this national infrastructure at a realistic scale.

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